

MorphoXAI: Anatomically Explainable AI for Facial Phenotyping toward Explainable Face2Gene Diagnosis



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A dissertation submitted for the degree of
Master of Science in Health Data Science

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August, 2025

Declaration

I declare that the work presented in this dissertation is, to the best of my knowledge and belief, original and my own work. The material has not been submitted, either in whole or in part, for a degree at this, or any other university.

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Abstract

Deep learning has achieved high performance in automated facial phenotyping, while clinical translation remains ambiguous and not clearly defined by opaque model behaviour and explanations that rarely align with clinical reasoning. This dissertation addresses that gap by introducing MorphoXAI, a framework that converts post-hoc attributions into anatomy-aligned semantic vectors by projecting model saliency onto 18 predefined craniofacial regions of interest (ROIs). Quantifying where decision evidence concentrates in anatomical terms enables region-wise statistical testing, stability assessment, calibration-aware interpretation, and governance-ready reporting, moving beyond anecdotal heatmaps toward auditable scientific evidence.

The framework was validated in two complementary settings. In a foundational Face2Gender task, semantic vectors consistently emphasised mandibular/mental and nasal-bridge regions for male faces versus periocular/forehead and zygomatic patterns for female faces, reflecting descriptors used in dysmorphology and facial-attribute analysis. In a clinical scenario, aligned with hypomimia, which is a condition of Parkinson’s Disease (PD), a post-intervention redistribution of decision weight toward lower-face regions (perioral/zygomatic/mandibular) was detected, consistent with improved facial expressivity and suggesting a potential quantitative biomarker of therapeutic response. Across both settings, projection into anatomical space preserved class-discriminative information while revealing coherent, physiologically plausible attentional archetypes that are difficult to extract from raw saliency maps alone.

A direct relationship between interpretability and calibrated operating regimes was stabilised because decision thresholds affect the way explanatory patterns behave; reliability diagrams and summary calibration errors helped determine the optimal threshold before

running the analysis. Stability audits spanning attribution families (gradient-based and perturbation-based), preprocessing, resolution, and layer choice mitigated known failure modes of saliency and framed explanation as an empirical design decision. Region-level evidence further supported fairness and robustness assessment through worst-group diagnostics and mechanistic audits of dataset shift (e.g., pose, illumination, background).

The principal contribution is an anatomy-aware, calibration-aware, and governance-aligned pipeline for explainable facial AI. Real-world effect of this system depends on external validation across multiple centres while strengthening uncertainty estimation methods and deploying the system through institutions that focus on safety and equity alongside accountability measures.

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Chapter 1

Introduction

1.1 Diagnostic odyssey & computational phenotyping

The diagnostic journey for individuals with rare genetic diseases is commonly lengthy and fragmented, basically an arduous “diagnostic odyssey” that may involve years of referrals and costly investigations before a definitive answer is reached (Salah and El-Jaick, 2019; Haendel and Vasilevsky, 2020). Many of these disorders, particularly neurodevelopmental syndromes, show *characteristic facial dysmorphologies* that trained dysmorphologists use to refine differential diagnosis (Pantelis et al., 2021). Yet, the total number of rare conditions and their phenotypic overlap imposes intrinsic constraints on human visual expertise and on the equitable availability of that expertise across care settings (Boycott, Rath, and Chong, 2017).

Deep learning has transformed facial phenotyping from tacit clinical know-how into a quantifiable, data-driven procedure. The landmark *DeepGestalt* study demonstrated that a convolutional neural network (CNN) trained on clinical images can rank candidate syndromes with high top- K accuracy, at times outperforming expert panels (Gurovich et al., 2019). Later prospective and utility evaluations confirmed that AI-suggested differentials can prioritise the diagnostic work-up and help *shorten time to diagnosis*, particularly in settings with limited specialist access (Ferry et al., 2022; Al-Saif et al., 2023). These developments position automated facial analysis as a decision-support technology in medical genetics—provided that challenges of transparency, calibration, and fairness are addressed (Rajpurkar et al., 2022; Kelly et al., 2019).

1.2 The “black box” dilemma

Models with strong discrimination tend to remain opaque because their non-linear hierarchical representations are difficult to scrutinise, which makes it challenging to hold them accountable in high-stakes contexts (McCoy, Bles, and Buanz, 2019). Lack of transparency is not a philosophical problem but a *safety and ethics* concern: opaque systems can conceal spurious shortcuts and demographic performance gaps, thereby risking inequitable care if unaudited (Geirhos, 2020; Mehrabi, Morstatter, and Saxena, 2021; Larrazabal, 2020; Cirillo, 2020). At the same time, clinicians demand explanations that can be checked, debated, and *aligned with clinical reasoning*, rather than post-hoc visuals that are difficult to interpret consistently (Kelly et al., 2019; Price, Gerke, and Cohen, 2022). Emerging governance efforts— within health systems and in broader AI policy—have therefore emphasised explainability, robustness, and documentation as preconditions for responsible deployment (Rajkomar, Dean, and Kohane, 2019; Vayena, Blasimme, and Cohen, 2018; Jobin, Ienca, and Vayena, 2019).

Two additional requirements are central. **Reliability and calibration.** Raw model scores are often miscalibrated, such that nominal confidences misrepresent empirical correctness. Clinical safety demands calibration assessment and transparent selection of operating thresholds to balance precision–recall trade-offs in an imbalanced, multi-syndrome setting (Murphy et al., 2021; Chan, 2020; Y.-L. e. a. Chen, 2023). **Governance and reporting.** Community artefacts such as data sheets set out clear statements of intended use, limitations, and fairness audits; regional guidance (e.g., EU trustworthy-AI efforts) similarly stresses documentation and risk management (Mitchell, 2019; Gebru and Morgenstern, 2021; Arnold, 2019).

1.3 Explainable AI in medical imaging

Explainable AI (XAI) has been developed to mitigate mitigate opacity by producing post-hoc rationales (Arrieta and Diaz-Rodriguez, 2020; Gunning, 2019). In medical imaging, saliency maps predominate. Gradient-based approaches such as Grad-CAM localise class-relevant regions via backpropagated gradients (Ramprasaath R Selvaraju and M. e. a. Cogswell, 2017; Chattopadhyay, 2018); perturbation-based methods like Score-CAM estimate class-score

changes under masked inputs (H. Wang and Z.-h.-i. e. a. Wang, 2020). Surveys document their widespread use across clinical applications (Islam, 2021; Van der Velden, 2022; Chaddad, 2023; Mishra, 2023).

Nevertheless, accumulating evidence points to faithfulness and stability concerns: saliency may persist after network randomisation, flip under benign preprocessing, or track priors rather than learned evidence (Adebayo, J. Gilmer, and Muelly, 2018; Hooker and Erhan, 2019; Kindermans, 2019; Ancona and Ceolini, 2017). Conceptual language of clinical anatomy does not align well with pixel heatmaps, even when they remain faithful. The analysis of dysmorphology depends on specific constructs including "depressed nasal bridge" and "smooth philtrum" and "mandibular hypoplasia" rather than pixel intensity distributions. As a result, heatmaps can be qualitatively illustrative, but quantitatively weak, impeding statistical comparisons across cohorts, operating points, or error types (Holzinger and Langs, 2019).

A complementary strand focuses on *concept-level* explanations. Techniques like TCAV, concept bottleneck models, regression concept vectors and network dissection enable hypothesis testing over concept scores by quantifying evidence along human-aligned factors (Kim, Wattenberg, and J. e. a. Gilmer, 2018; P. W. e. a. Chen, 2020; Graziani, 2020; Bau and Zhou, 2017; Ghorbani, 2019). Despite the promise, concept-level XAI has been involved in a series of limitations in application to craniofacial anatomy in genetics, where clinical reasoning is explicitly region-based.

1.4 Reliability and calibration

In diagnostic support, *how* a system is used is as critical as how well it discriminates. Best practice needs reporting reliability diagrams together with summary calibration errors and decision thresholds, which should be selected from the validation set based on clinical objectives (e.g., precision–recall trade-offs, positive predictive value restriction) (Murphy et al., 2021; Chan, 2020; Y.-L. e. a. Chen, 2023). It is almost mandatory that interpretability analyses should be performed at calibrated thresholds because explanation patterns depend on the operating point, thus preventing post-hoc cherry-picking and ensuring explanatory claims remain decision-relevant.

1.5 Fairness, dataset shift, and governance

Face-based models are sensitive to demographic composition and acquisition conditions. Documented differences and disparities need to be tested by bias audits and shift diagnostics (Buolamwini and Gebru, 2018; Larrazabal, 2020; Cirillo, 2020; Mehrabi, Morstatter, and Saxena, 2021). Robustness papers demonstrate that distribution shifts can cause significant performance weakness and modify patterns, which require external validity evaluations and detailed interpretation (Geirhos, 2020; D’Amour, 2020; Taori, 2020; Koh, 2021; Hendrycks and Dietterich, 2019). To support responsible translation, transparent documentation—model cards, data sheets, and associated governance practices—should accompany performance and interpretability reporting (Mitchell, 2019; Gebru and Morgenstern, 2021; Arnold, 2019).

1.6 MorphoXAI: Anatomically Explainable AI for facial phenotypes

This dissertation introduces *MorphoXAI*, a framework that converts pixel-level saliency into *quantitative, anatomy-aligned semantic vectors*. The pipeline comprises three stages:

Stage I — Activation generation. Class-specific maps are obtained with complementary saliency families (e.g., Grad-CAM, Score-CAM) to reduce method-specific artefacts (Ramprasaath R Selvaraju and M. e. a. Cogswell, 2017; H. Wang and Z.-h.-i. e. a. Wang, 2020).

Stage II — Anatomical projection. The face is partitioned into 18 craniofacial regions of interest (ROIs)—including brow complex, periorbital, nasal bridge, philtrum, lips, chin, and mandible—followed by aggregation of saliency within each ROI to form a *semantic activation vector*.

Stage III — Statistical and stability analysis. The resulting vectors are subjected to region-level hypothesis tests (with multiple-testing control), effect-size estimation, dimensionality reduction, clustering of attentional archetypes, and *error-signature* analysis (e.g., contrasts between true and false predictions). All analyses were executed at calibration operating points to warrant the decision relevance and robustness (Chan, 2020; Y.-L. e. a. Chen, 2023).

MorphoXAI enables evidence-based explanations by linking model evidence to clinical anatomy, which transforms unreliable information into auditable evidence that supports structured comparisons between cohorts and conditions.

1.7 Facial expressivity in Parkinson’s disease

Beyond genetic dysmorphology, facial appearance also reflects neurological states. *Hypomimia* (facial bradykinesia) is one of them, and well-known as a main feature of Parkinson’s disease (PD) with functional and social impact (Scherpf, 2019). Recent machine-learning work has explored PD detection and severity assessment from facial dynamics (Abreu, 2021; Tolosana, 2022). The application of an anatomy-aligned interpretability pipeline in this context provides a challenging evaluation of generalisability and physiological validity to determine if ROI vectors detect clinically relevant changes between upper- and lower-face areas under various clinical scenarios.

1.8 Research gap and questions

The literature establishes accurate AI-assisted facial phenotyping (Gurovich et al., 2019; Ferry et al., 2022), widespread yet *problematic* reliance on pixel-space saliency (Adebayo, J. Gilmer, and Muelly, 2018; Hooker and Erhan, 2019; Kindermans, 2019), the promise of concept-level approaches (Kim, Wattenberg, and J. e. a. Gilmer, 2018; P. W. e. a. Chen, 2020; Graziani, 2020), and the necessity of calibration, fairness, and documentation for clinical translation (Rajpurkar et al., 2022; Mitchell, 2019; Gebru and Morgenstern, 2021). What remains missing is a unified, anatomy-grounded, quantitatively testable explainability pipeline for facial phenotypes that operates at clinically chosen thresholds and supports governance-ready reporting. Accordingly, the following questions are addressed:

- RQ1.** Can post-hoc activation maps be transformed into *ROI-based semantic vectors* that preserve class-discriminative information while enabling region-level hypothesis testing?
- RQ2.** Do ROI-based semantic vectors reveal *consistent, statistically significant* anatomical patterns across classification tasks and clinical cohorts, beyond what simple heatmaps suggest?

RQ3. How do *calibration choices and operating thresholds* affect both performance metrics and the *stability* of region-level interpretability conclusions?

RQ4. Under *dataset shift and demographic imbalance*, are ROI vectors more robust—or at least more *auditable*—than pixel saliency for bias and fairness analyses?

1.9 Contributions

1. **Anatomy-aware explainability.** Projection of saliency into *craniofacial ROI vectors* that align model evidence with clinical concepts and enable region-level statistical testing and error-signature analysis (Ramprasaath R Selvaraju and M. e. a. Cogswell, 2017; H. Wang and Z.-h.-i. e. a. Wang, 2020; Kim, Wattenberg, and J. e. a. Gilmer, 2018; P. W. e. a. Chen, 2020).
2. **Calibration-aware interpretation.** Evaluation at *clinically chosen operating points* with reliability diagrams and calibration summaries, and propagation of these choices into interpretability analyses (Murphy et al., 2021; Chan, 2020; Y.-L. e. a. Chen, 2023).
3. **Cross-domain evidence.** Demonstrates evidence across two different environments, *Face2Gender* (gender classification) and *Parkinson’s cohorts* (DBS classification) to evaluate the generalisability and physiological validity of anatomy-aligned evidence (Pantelis et al., 2021; Scherpf, 2019; Abreu, 2021; Tolosana, 2022).
4. **Governance-ready reporting.** Integration of model-card and data-sheet style documentation, with explicit attention to fairness and shift considerations (Mitchell, 2019; Gebru and Morgenstern, 2021; Arnold, 2019).

1.10 Chapter outline

- **Chapter 2 (Methods).** Datasets and preprocessing; baseline architectures and training; CAM generation (Grad-CAM/Score-CAM); MediaPipe landmarks and the 18-ROI anatomical atlas; semantic projection from heatmaps to ROI vectors; statistical design (Mann–Whitney with BH–FDR, two-way ANOVA, multivariate logistic models

and GEE), dimensionality reduction and clustering; robustness audits (seed/split, blur stress tests); calibration (temperature scaling, ECE/Brier, operating-point selection).

- **Chapter 3 (Results).** **Face2Gender (Phase 1)** includes discrimination metrics, calibration and thresholding, region-level contrasts by sex/age, stability and robustness findings, While **DBS (Phase 2)** focused on treatment-response discrimination and calibrated operating point, redistribution of attention across ROIs, univariate and multivariate effects with GEE sensitivity, attentional archetypes, and robustness checks.
- **Chapter 4 (Discussion).** Synthesis and positioning of MorphoXAI; interpretation of principal findings; faithfulness and stability of explanations; generalisation, dataset shift, and fairness; clinical implications and governance (documentation, auditing, calibration in practice); limitations and directions for future work.
- **Chapter 5 (Conclusion).** Main contributions of MorphoXAI, calibrated, auditable explanations—key takeaways for clinical translation, and concrete next steps for responsible deployment.
- **Appendices.** Model/data cards, additional figures and tables.

Chapter 2

Methods

This chapter presents a complete and technical description of the methodological framework developed and employed in this dissertation. This section provides an overview of the MorphoXAI framework, which constitutes the main contribution of this paper. Thereafter, the chapter describes the experimental approaches of two progressive phases with datasets, models and analytical methods which were employed during gender classification validation (Phase 1) and facial motor recovery assessment for Parkinson’s disease patients after DBS (Phase 2). The objective is to deliver an exact and detailed explanation of the entire process, starting from MorphoXAI development through gender and DBS analysis.

2.1 MorphoXAI Framework

The MorphoXAI framework was designed to fundamentally address the ”black-box” problem in deep learning-based facial analysis. Its primary objective is to transform opaque, pixel-level saliency maps, which indicate where a model is looking, into quantitative, anatomically-grounded semantic vectors, which explain what anatomical structures the model is focusing on and how much. This transition from a qualitative visualisation to a quantitative feature vector is the key innovation that enables rigorous statistical analysis of the model’s internal reasoning (Adebayo, J. Gilmer, and Muelly, 2018; Hooker and Erhan, 2019).

MorphoXAI operates via a three-stage pipeline, which was applied consistently across both experimental phases to ensure comparability. Figure 2.1 gives an overview of this process, showing how an original image is processed through activation map generation

and anatomical segmentation to produce a final and easily interpretable semantic activation vector.

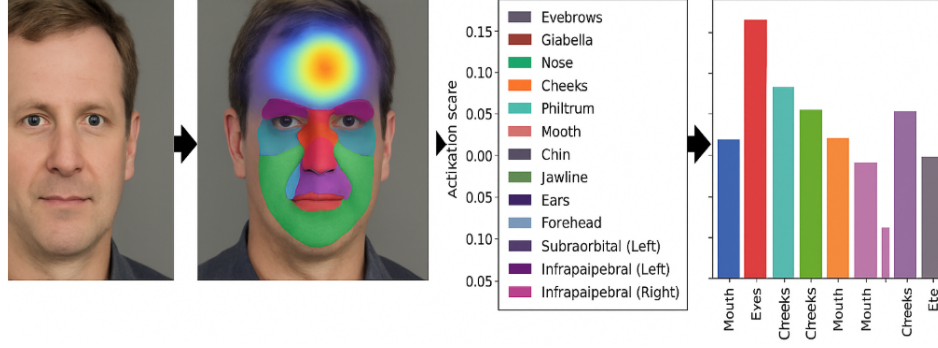


Figure 2.1: MorphoXAI pipeline

2.1.1 Class Activation Map (CAM) Generation

The first stage requires the creation of spatial attention heatmaps from input image x and target class score y . Two different CAM methods were chosen to provide a complete analysis of the results. The use of both methods enables essential validation of the obtained explanations. In each Phase, the last convolutional block is targeted, because this block contains the most significant semantic information (Ramprasaath R. Selvaraju, M. Cogswell, and Das, 2017; H. Wang, Z. Wang, and Du, 2020).

- **Grad-CAM:** This method calculates an importance weight, α_k , for each feature map, A^k , by global-average-pooling the gradients of the class score, y , with respect to that map's pixels (i, j) . These weights are then used for a linear combination of the feature maps, and a *ReLU* function is applied to isolate positive contributions, resulting the final heatmap, $L_{\text{Grad-CAM}}$. This process is formally defined by the equation below (Ramprasaath R. Selvaraju, M. Cogswell, and Das, 2017).

$$\alpha_k = \frac{1}{Z} \sum_{i,j} \frac{\partial y}{\partial A_{ij}^k}, \quad L_{\text{Grad-CAM}} = \text{ReLU} \left(\sum_k \alpha_k A^k \right) \quad (2.1)$$

- **Score-CAM:** This is a gradient-free method that computes its weights, w_k , through a series of forward passes. The proportionality of the weight is determined by using each

feature map, $\hat{A}^k$, to mask the input image, x , and measuring the resulting model score, $f(\cdot)$. The final heatmap is then constructed as a weighted sum of the feature maps. This relationship is expressed as follows (H. Wang, Z. Wang, and Du, 2020).

$$w_k \propto f\left(x \odot \hat{A}^k\right), \quad L_{\text{Score-CAM}} = \text{ReLU}\left(\sum_k w_k \hat{A}^k\right) \quad (2.2)$$

2.1.2 Anatomical Segmentation

To ground the explanations in a consistent biological context, the face is programmatically segmented into anatomically significant ROIs using the MediaPipe Face Mesh library (Lugaresi, 2019; Kartynnik and al., 2019). In Phase 1, the base model detecting 468 landmarks was used. In Phase 2, MediaPipe Face Mesh was improved with iris refinement enabled (refine_landmarks=True), yielding 478 landmarks for more precise centroids per ROI (Bazarevsky and al., 2020). The anatomical atlas used in both phases consists of 18 Regions of Interest (ROIs), programmatically defined from the MediaPipe Face Mesh landmarks. To ensure robustness, soft Gaussian masks were used, defined by the formula:

$$M_r(p) = \exp\left(-\frac{\|p - \mu_r\|^2}{2\sigma_r^2}\right) \quad (2.3)$$

where p is a pixel coordinate in the image, μ_r is the centroid coordinate of the landmarks for region r , and σ_r is the region-specific standard deviation that controls the size of the mask (Newell, Yang, and J. Deng, 2016).

The process depends on the region-specific standard deviation (σ_r), which determines both the size and decay of each Gaussian mask to establish the spatial boundaries of each anatomical ROI. A fixed sigma for all regions would be anatomically simple, as features like the forehead are substantially larger than the eye.

Therefore, a heuristic approach was adopted where σ_r was set to be proportional to the anatomical size and expressive range of the underlying landmark group. Regions spanning a larger area (e.g., forehead, left jaw) were assigned a larger sigma, while more compact regions (e.g., left eye, right eyebrow) received a smaller sigma. This makes sure that the resulting soft masks are morphologically realistic, providing a biologically representation of

facial regions while considering the overlapping influence of expressive muscles (Ekman and Friesen, 1978). The specific sigma values along with the created regions, refined through empirical observation to ensure a correct coverage without overlapping, are detailed in Table 2.1.

Region	Sigma (σ_r)	Anatomical Rationale
forehead	20	Largest, broad region
left_jaw	15	Large, extensive region
right_jaw	15	Large, extensive region
lips	14	Large, highly expressive region
left_cheek	13	Broad, mid-sized region
right_cheek	13	Broad, mid-sized region
nose	13	Prominent, centrally located region
chin_center	11	Mid-sized central feature
left_cheekbone	10	Compact bone structure
right_cheekbone	10	Compact bone structure
chin_left	10	Smaller, lateral chin area
chin_right	10	Smaller, lateral chin area
left_temple	10	Smaller, lateral region
right_temple	10	Smaller, lateral region
left_eyebrow	8	Small, defined feature
right_eyebrow	8	Small, defined feature
left_eye	7	Smallest, most compact region
right_eye	7	Smallest, most compact region

Table 2.1: **Heuristic Sigma (σ_r) Values for Gaussian Mask Generation**

The process shown in Figure 2.2 uses MediaPipe Face Mesh facial landmark detection to establish the borders of essential facial features including eyes, nose, lips and jaw which creates a standardised anatomical atlas for each face.

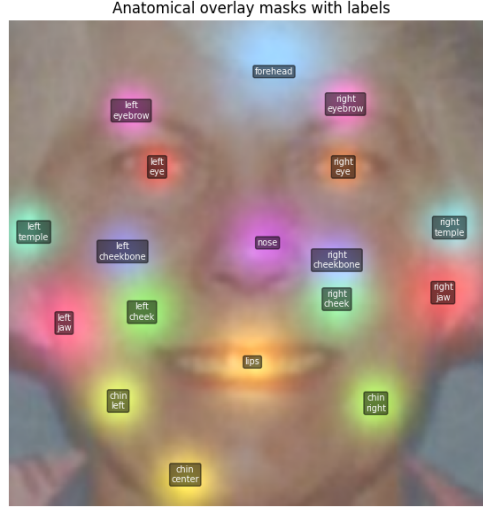


Figure 2.2: Visualisation of the anatomical atlas

2.1.3 Semantic Activation Vector Quantification

Each CAM, L , start going through a *ReLU* function and then min-max normalised on a per-image basis to ensure comparability across different images. The normalised map, L' , is resized to 224x224 pixels. This process is defined as:

$$L' = \frac{L - \min(L)}{\max(L) - \min(L) + \varepsilon}, \quad L' \in [0, 1] \quad (2.4)$$

This normalised map is then projected onto the atlas of soft Gaussian ROI masks (Figure 2.3), M_r .

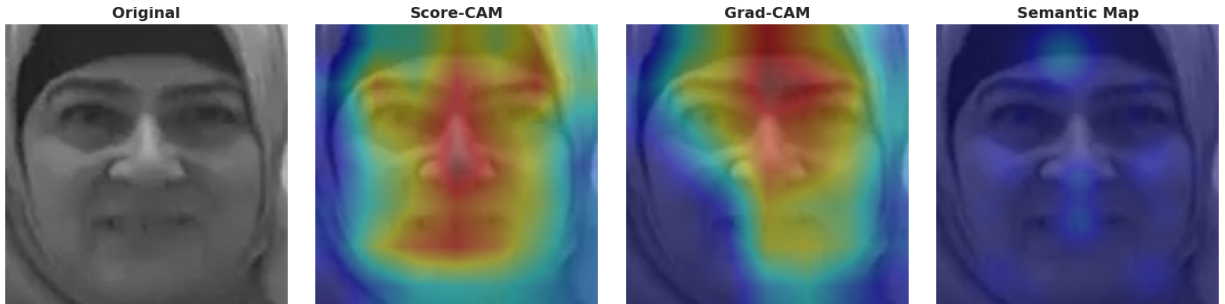


Figure 2.3: Semantic conversion process.

The regional attention share, s_r , is calculated as the sum of the CAM’s energy within the mask divided by the total energy of the CAM. The remaining CAM energy not captured by any of the defined anatomical masks is aggregated into an ‘other’ category. This ensures that 100% of the model’s attention is accounted for in the final semantic vector, but is omitted in some figures to avoid plotting distortion.

$$s_r = \frac{\sum_{i,j} L'(i,j)M_r(i,j)}{\sum_{i,j} L'(i,j)}, \quad s_{\text{other}} = 1 - \sum_r s_r \quad (2.5)$$

2.2 Phase 1: Foundational Validation (Face2Gender)

The first step involved building and validating the MorphoXAI framework through gender classification using a large publicly available dataset. The main objective was to create a testing framework which would validate the framework’s ability to explain model reasoning on a basic facial analysis task before advancing to a more sophisticated clinical application.

2.2.1 Dataset

In this first step, the UTKFace dataset was utilised, a public-access collection of over 20,000 facial images. This dataset was selected as the ideal testbed for three primary reasons:

1. **Size (large)**, which allows for robust training and statistical analysis.
2. **Richness**, because the filename is encoded with demographic labels (age, gender, ethnicity).
3. **Uncontrolled nature** (varied poses, lighting, and expressions), which ensures that models validated on it are robust to real-world variability (Z. Zhang, Song, and Qi, 2017).

The data was systematically selected based on the following parameters. The filename of each image was analysed to extract the age and gender labels (e.g., [age]-[gender]-...jpg). Secondly, a filter was implemented to choose only images with an age of at least 10 years (min_age=10), a decision intended for concentrating the investigation on post-pubescent and adult facial morphology, when craniofacial gender variation is more physiologically evident.

The final set of images was split into training ($\approx 80\%$) and test ($\approx 20\%$) using an age-group-stratified procedure (seed = 42). To preserve minority representation, we enforced a minimum of 400 Teen images in the evaluation split. This stratification ensured that the held-out set was demographically representative rather than skewed toward any age range.

2.2.2 Preprocessing and Data Augmentation

A standardised preprocessing pipeline was applied to all images. This pipeline includes image resizing to 224×224 and transformation to RGB. Then the tensors were normalised to $[-1, 1]$ per channel using $\mu = 0.5$ and $\sigma = 0.5$. During the whole training process, a fixed batch size of 64 was used.

To enhance model generalisation and prevent overfitting, a data augmentation was applied specifically to the training set. This included random horizontal flipping to ensure the model does not learn a left/right facial bias; random rotations ($\pm 10^\circ$); and colour jittering to vary brightness and contrast. These transformations create a more diverse training set, encouraging the model to learn robust features that are invariant to minor changes in orientation and lighting conditions.

2.2.3 Model Architecture and Training

To determine a strong baseline for the XAI analysis, a VGG16 convolutional neural network was employed, leveraging the principles of transfer learning. The model was initialised with weights pre-trained on the ImageNet dataset (Simonyan and Zisserman, 2015; J. Deng, Dong, and Socher, 2009). This approach was chosen to control the rich, low-level feature detectors (e.g., edges, textures, shapes) learned from a big and diverse dataset. The convolutional base of the VGG16 model was frozen for all its parameters. Consequently, these layers acted as a fixed feature extractor, preventing their weights from being updated during training.

The original classifier head was replaced with a new, randomly initialised sequence of layers tailored for the binary gender classification task. Only the parameters of this new classifier were trained. This practice used to ensure that the model’s predictive capabilities are adapted specifically to the UTKFace dataset, while grounding the analysis in a powerful, pre-existing feature space. This avoids the computational expense and potential instability of training

a deep network from scratch and provides a solid foundation for interpreting which of these general-purpose features the model deems relevant for gender discrimination in faces.

The model was trained for 30 epochs using the Adam optimiser ($\text{lr} = 10^{-4}$, weight decay $= 10^{-5}$) with Binary Cross-Entropy Loss (Kingma and Ba, 2015). A ReduceLROnPlateau scheduler (factor = 0.5, patience = 2) was stepped based on the loss evaluated on the held-out test split used as a validation proxy at the end of each epoch. The best checkpoint was selected by held-out accuracy, and all metrics are reported on that same split acknowledging some optimistic estimates; to mitigate optimism from this protocol, we additionally report ROC-AUC and the confusion matrix to characterise robustness. therefore, in this phase results are interpreted as a methodological validation (stress-testing the XAI pipeline) rather than as definitive generalisation claims like phase 2.

The model’s final classifier layer uses a Sigmoid activation function to identify the network’s logit outputs (z_i) to a probability score, $\hat{p}_i = \sigma(z_i)$, which ranges from 0 to 1. The training was regulated by a Binary Cross-Entropy Loss (BCELoss) criterion, which quantifies the penalty by comparing this predicted probability $\hat{p}_i$ to the true binary label y_i

This loss function calculates prediction penalties by the comparison of the output probability $\hat{p}_i$ to the true binary label y_i . The total loss across a batch of images is calculated as the mean of the individual cross-entropy values by the following formula:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{p}_i) + (1 - y_i) \cdot \log(1 - \hat{p}_i)] \quad (2.6)$$

2.3 Phase 2: Clinical Application (MorphoXAI-DBS)

The last phase aimed to test the clinical effectiveness of the validated model by using it to address a complex clinical problem: assessing facial motor changes in Parkinson’s disease (PD) patients who underwent DBS therapy.

2.3.1 Dataset

This study analysed a public collection of facial video frames from 48 patients with Parkinson’s disease extracted from a YouTube video. Labels were obtained from the image

filenames, where files containing `left_` were assigned the Pre-DBS state (class 0) and files containing `right_` were assigned the Post-DBS state (class 1). A strict patient-wise stratified split was enforced, splitting the 48 patients where training (32 patients, $\approx 70\%$), validation (8 patients, $\approx 15\%$), and test (8 patients, $\approx 15\%$) sets contain independent patients. This methodology is critical to prevent data leakage and ensures that the model is evaluated on its ability to generalise to entirely unseen patients, not merely unseen images of patients present in the training set. All training and evaluation used a batch size of 32.

2.3.2 Preprocessing and Data Augmentation

Images were converted to 3-channel grayscale, resized to the experimental resolution, transformed to tensors, and normalised with ImageNet statistics ($\mu=[0.485,0.456,0.406]$, $\sigma=[0.229,0.224,0.225]$). This grayscale conversion prompts the network to focus specifically on luminance information, such as shape, contour, and texture, which are directly relevant to the clinical signs of hypomimia. Augmentations for the training set included `RandomHorizontalFlip` and `RandomAffine` transformations ($\pm 15^\circ$ rotation, $\pm 10\%$ translation, 0.9–1.1 scale, $\pm 10^\circ$ shear) with light `ColorJitter`.

2.3.3 Model Architecture and Training

To empirically determine the optimal model for this subtle clinical task, a systematic grid search was performed across 20 experimental setups (5 model architectures $\times$ 4 image resolutions).

The Training Protocol was executed by a precise two-phase transfer learning. During the first 15 epochs, the model trained only its final classification head (Adam, $\text{lr}=10^{-3}$) while keeping the pretrained backbone fixed from ImageNet. This rapidly adapts the model to the new task. Second, the entire network was unfrozen and fine-tuned for 30 epochs with a much lower learning rate (Adam, $\text{lr}=10^{-5}$). A `ReduceLROnPlateau` tracked validation accuracy to decrease the learning rate by 0.2 after three consecutive epochs without improvement. The chosen patience duration of three epochs served to achieve equilibrium between fast learning and stability because it provided the model enough time to address minor performance drops before confirming a learning plateau required a learning rate decrease. The model checkpoint with the best validation accuracy was saved for the final evaluation.

In this phase, the model head does not include a final activation function. BCEWithLogitLoss, which provides a sigmoid function within a numerically stable logistic loss calculation (Goodfellow, Bengio, and Courville, 2016). This function calculates the loss across a group of images by penalising the model according to the distance between the predicted logit, z_i , and the true binary label, y_i , as you can see in the formula below.

$$\mathcal{L}_{\text{BCE-logits}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \sigma(z_i) + (1 - y_i) \log (1 - \sigma(z_i))], \quad \sigma(z) = \frac{1}{1 + e^{-z}} \quad (2.7)$$

2.3.4 Decision Thresholding and Metrics

Following model selection (best validation accuracy at a fixed 0.5 operating point), the logits on the validation set were calibrated and then selected an operating threshold τ^* by sweeping $\tau \in \{0.01, \dots, 0.99\}$ over the *calibrated* validation probabilities to maximize validation accuracy. The chosen τ^* was applied once to the independent test set, as you can see in the formula below.

$$\tau^* = \arg \max_{\tau \in \{0.01, \dots, 0.99\}} \frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \mathbf{1}(\mathbf{1}\{\hat{p}_i \geq \tau\} = y_i). \quad (2.8)$$

The optimal threshold τ^* was then applied once to compute threshold-dependent test metrics (Accuracy, class-wise F1) and the confusion matrix. In parallel, the Precision–Recall (PR) curve and Average Precision (AP) were computed from the continuous scores $\{\hat{p}_i\}$ (i.e., threshold-free metrics) (Davis and Goadrich, 2006). For visualisation, the PR curve was annotated at the operating point closest to τ^* by selecting $j^* = \arg \min_j |\text{thresholds}[j] - \tau^*|$ and plotting $(\text{recall}[j^* + 1], \text{precision}[j^* + 1])$, following scikit-learn’s convention that `precision/recall` have length `|thresholds| + 1`.

2.3.5 Calibration of Probabilistic Outputs

To improve the reliability of the predicted probabilities, a post-hoc calibration was applied via temperature scaling. A single scalar temperature $T > 0$ is learned on the *validation* set

by minimizing the binary cross-entropy (LBFGS optimiser) (Guo et al., 2017). Given logits z , calibrated probabilities are $\hat{p} = \sigma(z/T)$. The learned T is then fixed and applied to both validation and test.

Calibration quality was quantified with the Expected Calibration Error (ECE, 15 bins) and the Brier score, and visualised using a reliability diagram (Naeini, Cooper, and Hauskrecht, 2015; Brier, 1950). A 95% bootstrap CIs (1,000 resample) for Accuracy and Average Precision was also reported.

2.4 Statistical Design and Inference

The following section presents a complete quantitative analysis of the 18-dimensional semantic activation vectors obtained from MorphoXAI framework. The statistical analysis has two purposes: to detect meaningful variations and to establish the functional importance of model-derived attentional patterns while measuring their predictive power, structural properties and assessing their robustness to changes. All statistical tests were conducted with a significance level of $\alpha=0.05$.

2.4.1 Group Comparisons and Interaction Effects

To test if the model’s attentional patterns changes significantly between the primary experimental groups, a two-sided Mann-Whitney U test was applied to the activation scores for each of the 18 ROIs in a separate way (Mann and Whitney, 1947). To control for the risk of false positives emerging from multiple comparisons across the 18 ROIs, the p-values were adjusted using the Benjamini-Hochberg False Discovery Rate (FDR) correction (Benjamini and Hochberg, 1995). This test was selected for its robustness, as it tests the null hypothesis that the distributions of attention scores for a given ROI are the same between two independent groups, without assuming that the data follows a normal distribution—a condition often ignored by derived metrics like attention scores.

To support this statistical test, boxplots were developed to deliver a visual representation of these distributions. These figures give a visual comparison of the median, interquartile range, and outliers of the model’s attention for each region between the experimental groups, making the results of the Mann-Whitney U test easier to interpret.

To investigate more complex relationships, a two-way Analysis of Variance (ANOVA) was employed in both phases to test for a statistical interaction between the experimental factors (Fisher, 1925). A significant interaction term validates the hypothesis that a demographic difference or treatment effect manifests as a redistribution of attention across the face.

- In Phase 1, the model examined the interaction between Gender and AgeGroup to see if gender-specific attention patterns changed across different ages.
- In Phase 2, the model examined the interaction between the clinical Group (Pre/Post-DBS) and the anatomical Region to see whether the impact of DBS on attention varied across different facial regions.

The model for this analysis is specified as follows:

$$s_r \sim \text{Group} + \text{Region} + \text{Group} : \text{Region} \quad (2.9)$$

2.4.2 Analysis of Attentional Patterns in Correct vs. Incorrect Predictions

In both phases, to validate that the model’s learned attentional patterns were functionally relevant to its performance, an analysis was conducted to compare regional attention between correctly and incorrectly classified images. This analysis functions as an internal validation of the model’s logic to verify that the XAI-identified features are both correlational and operationally critical for successful prediction..

The methodology involved partitioning the test set into two subsets: one containing all correctly classified images (True Positives and True Negatives) and another containing all incorrectly classified images (False Positives and False Negatives). For each of the 18 ROIs, the mean semantic activation score (s_r) was computed independently for each subset. The main evaluation criterion for this study was the difference in mean activation. A positive value indicates that higher attention on a specific region is associated with successful classification, while a negative value suggests that attention on that region may be linked to prediction errors.

2.4.3 Logistic Modelling of Semantic Vectors

To go further group comparisons and quantify the predictive relationship between the model's attentional patterns and its classification output, a series of logistic regression models were developed. The primary objective of this analysis was to determine how a marginal increase in attention allocated to specific facial regions affects the probability of a positive class prediction ($y=1$). This was achieved by modelling the log-odds of the classification outcome as a function of the 18-dimensional semantic activation vectors (s). The analysis was organised in three stages, starting with an initial univariate testing to evaluate individual regional effects (Only Phase 2), followed by a multivariate model to assess combined and adjusted effects, and ending with a final sensitivity analysis using Generalised Estimating Equations (GEE) to validate the findings in clustered data settings.

2.4.3.1 Univariate logistic regression

The first step involved running a separate univariate logistic regression models for all 18 anatomical Regions of Interest ($ROIs$). The analysis performed during Phase 2 served to screen individual regions for their strongest independent relationship with classification outcomes. For each region r , the model was specified as:

$$\text{logit } P(y = 1 \mid s_r) = \beta_0 + \beta_r s_r \quad (2.10)$$

The Odds Ratio ($OR_r = e^{\beta_r}$) was calculated by exponentiating the coefficient β_r . The value shows how the probability of positive classification changes when the regional attention share (s_r) increases by one unit while ignoring other facial region influences. This method helps identify regions with powerful, unaltered predictive power, although it remains susceptible to variable bias that may be overlooked.

2.4.3.2 Multivariate logistic regression

A multivariate logistic regression model was implemented to evaluate the independent effects of each region while accounting for their natural multicollinearity (Hosmer, Lemeshow, and Sturdivant, 2013). This model serves as the primary tool for interpreting the functional importance of the complete 18-dimensional semantic vector. In both phases, all 18 regional attention scores (s_r) were standardised to have a mean of zero and a standard deviation of

one prior to model fitting. This standardisation allows for the direct comparison of model coefficients.

However, the specific implementation of the model was adapted to the objectives of each phase and in a way to enhance phase 2 for clinical purposes, differing in the use of regularisation and the method for estimating confidence intervals.

For Phase 1 (Gender Classification), a standard multivariate logistic regression was employed. This model was implemented using the statsmodels. Logit function and did not include an explicit regularisation penalty, focusing on maximum likelihood estimation to identify the principal predictive features in the dataset.

For the clinical analysis in Phase 2 (DBS), a more complex, regularised model was used to enhance stability and prevent overfitting. This was a multivariate logistic regression with an L2 penalty (Ridge regularisation) (C=1.0), implemented via scikit-learn. Furthermore, to account for the clustered data structure where multiple image frames were derived from the same patient (Le Cessie and Van Houwelingen, 1992), confidence intervals were computed using a robust, non-parametric patient-level bootstrap procedure with 1,000 resamples.

For both models, the final predictive equation is specified as:

$$\text{logit } P(y = 1 \mid s) = \beta_0 + \sum_{r=1}^{18} \beta_r s_r, \quad \text{OR}_r = e^{\beta_r} \quad (2.11)$$

The resulting Odds Ratio for a given region, OR_r , is interpreted as the multiplicative change in the odds of a positive classification for a one-standard-deviation increase in attention on that region, holding the attention in all other 18 regions constant. While the predictive formula is identical, the coefficients (β_r) for each model are estimated in a different way due to the application of regularisation and the bootstrap procedure in Phase 2.

2.4.3.3 Sensitivity Analysis via Generalised Estimating Equations

To assess the findings from the multivariate model and to ensure their robustness to different statistical assumptions, in Phase 2, a Generalised Estimating Equations (GEE) model was implemented to test the sensitivity. GEE is a method designed specifically for analysing clustered or longitudinal data by modelling the population-averaged response (Liang and Zeger, 1986).

The model was specified with a binomial family, a logit link function, and an exchangeable correlation structure. The model assumes equal correlation between all images extracted from the same patient via an exchangeable correlation structure, which provides an efficient method to work with cluster dependencies. A key advantage of GEE is that it yields consistent coefficient estimates even if the working correlation structure is misspecified, and it produces standard errors that are robust to this potential misspecification. A high degree of concordance between the Odds Ratios derived from the L2-penalised bootstrap model and the GEE model provides strong evidence for the robustness of the identified attentional biomarkers.

2.4.4 Dimensionality Reduction and Unsupervised Clustering

This part of the analysis aimed to explore the global structure of the 18-dimensional "attention space" in a purely data-driven, unsupervised manner. As a prerequisite, the semantic vectors were first standardised using *StandardScaler* to ensure all regions contributed equally to the distance-based algorithms.

- In Phase 1, PCA and t-SNE were used for exploratory visualisation to assess whether the semantic vectors contained a separable demographic structure (Jolliffe, 2002; Maaten and Hinton, 2008).
- In Phase 2, the analysis was extended to include a formal partitioning algorithm, K-Means clustering ($k=3$), with the explicit goal of discovering data-driven "attentional archetypes" that might correspond to the clinical states (Lloyd, 1982).

t-SNE achieves its non-linear visualisation by minimising the Kullback-Leibler (KL) divergence between the high-dimensional and low-dimensional probability distributions of data point neighbourhoods, defined by the objective function:

$$\mathcal{C}_{\text{t-SNE}} = \text{KL}(P||Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (2.12)$$

K-Means clustering partitions the data by finding centroids for k clusters that minimise the "inertia" or within-cluster sum of squares. Its objective is to find the set of cluster centroids,

μ_c , that minimises the total squared Euclidean distance from each data point, s_i , to its assigned cluster's centre, according to the function:

$$\min_{\{\mu_c\}, \{c_i\}} \sum_i \|s_i - \mu_{c_i}\|_2^2, \quad c \in \{1, \dots, k\}. \quad (2.13)$$

2.4.5 Robustness of Explanations

To ensure that the XAI explanations were stable and not just artefacts of high-frequency noise, a robustness test was conducted in both phases and XAI techniques (Grad-Cam and Score-Cam) by degrading the input images with a Gaussian blur. However, the specific methodology and quantification metric differed.

- In Phase 1, a moderate Gaussian blur ($\sigma=5$) was applied to the input images. To quantify the stability, the Pearson correlation coefficient (r) was employed between the original and blurred 18-dimensional semantic vectors. This coefficient evaluates linear vector relationships between the two vectors by showing that a value close to 1 means that the attention distribution across 18 ROIs stays highly consistent after perturbation.
- In Phase 2, a stronger Gaussian blur ($\sigma=10$) was used to represent a more significant perturbation. Stability was quantified by calculating the mean absolute change (Δ) between the original normalised CAM (L'_{orig}) and the blurred CAM (L'_{blur}) over the image domain Ω . This provides a pixel-level quantification of the explanation's stability, where a low value indicates a robust explanation that is not dependent on high-frequency image artefacts. The metric is formally defined as:

$$\Delta = \frac{1}{|\Omega|} \sum_{(i,j) \in \Omega} |L'_{orig}(i,j) - L'_{blur}(i,j)| \quad (2.14)$$

where a low value indicates a robust explanation. To complement this pixel-level metric, the stability of the final semantic vectors was also assessed by summarising the absolute change per region, $\delta = |s_r - s'_r|$, and by computing the Pearson correlation between the original and blurred vectors to confirm that the relative anatomical ranking of attention is preserved.

2.4.6 Computational Environment and Reproducibility

The entire data processing, together with model training and statistical analysis, took place within Python version 3.11.13 on Google Colab. Throughout the experiment, the versions of essential libraries were fixed to maintain stability and reproducibility of the results.

The deep learning environment used was PyTorch version 2.6.0+cu124, which integrates the torchvision library version 0.21.0+cu124 for model architectures and image transformations. All computational operations, which included model training and inference, were boosted through a CUDA-compatible graphics processing unit (GPU).

The fundamental libraries for data processing, scientific computation, and machine learning consist of pandas version 2.2.2, NumPy version 1.26.4 and scikit-learn version 1.x. The OpenCV (cv2) version 4.11.0 was used for executing computer vision activities, which included image processing for robustness analysis. The GEE analysis, together with all statistical modelling, was performed using the statsmodels package at version 0.14.5.

A global random seed of 42 was applied uniformly across all scripts to guarantee reproducibility of all stochastic elements in trials, including dataset splits, data augmentation, and model weight initialisation.

The complete source code for the MorphoXAI framework, including the implementation of the data processing pipelines, model training, and all analyses reported in this dissertation, is publicly available in a GitHub repository at:

- **Face2Gender (Phase 1):** https://github.com/fmiralles93/MorphoXAI/blob/main/morphoxai_vf.py
- **DBS (Phase 2):** [https://github.com/fmiralles93/DBS_MORPHOXAI/blob/main/dbs_v35\(full_en_2\).py](https://github.com/fmiralles93/DBS_MORPHOXAI/blob/main/dbs_v35(full_en_2).py)

Chapter 3

Results

This chapter presents the empirical findings from the two-phase experimental design detailed in Chapter 2. The results are organised to address the research questions presented in this dissertation. Section 3.1 validates the MorphoXAI framework (Phase 1) through training a binary gender classification model to demonstrate the framework’s ability to extract biologically valid and statistically significant explanations from a large dataset. Section 3.2 then presents the results of the clinical application (Phase 2), where the validated framework was deployed to a more complex and subtle problem: quantifying facial motor recovery in Parkinson’s disease patients post-DBS. This second phase functions as a proof-of-concept for the framework’s potential in clinical diagnostic settings to address the challenges associated with Face2Gene applications.

3.1 Phase 1: Foundational Validation on Gender Classification

3.1.1 Predictive Model Performance

The VGG16 convolutional neural network, which was trained and tested with the public UTKFace dataset, achieved a high precision at the binary gender classification task. Figure 3.1 shows that the model obtained a final accuracy of 92.34% and an Area Under the Curve (AUC) of 0.97 on the test set of 4,165 images. The confusion matrix shows that the two classes did almost the same, with 1,789 true positives and 2,057 true negatives. The high

AUC value shows that the classes are well-separated at all operating thresholds, making the model a good base for future interpretability testing utilising the MorphoXAI framework.

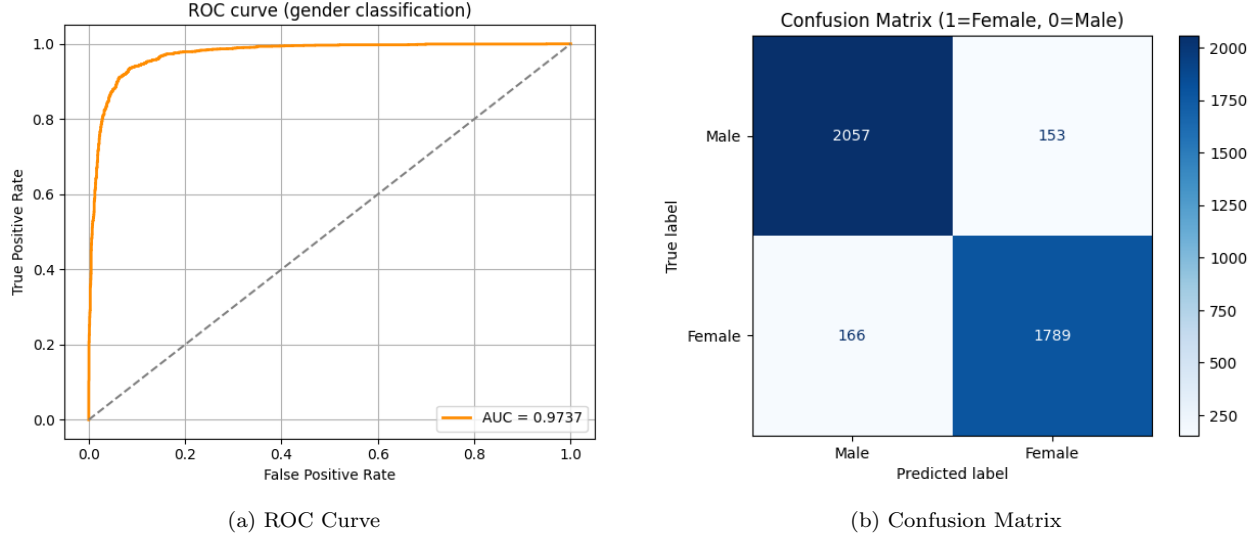


Figure 3.1: Model Performance in Gender Classification

3.1.2 Anatomical Attention Patterns Revealed by MorphoXAI

With the high predictive accuracy established for each model, MorphoXAI framework was used to reveal the process when the model is making the classifying decisions. A qualitative comparison of the most well-known Class Activation Mapping (CAM) techniques was employed to discover their relevance and convenience for generating precise anatomical explanations. These comparisons show that in this Phase, Score-CAM consistently produces more focused and interpretable heatmaps than Grad-CAM, which tends to be more diffuse (Figure 3.2).

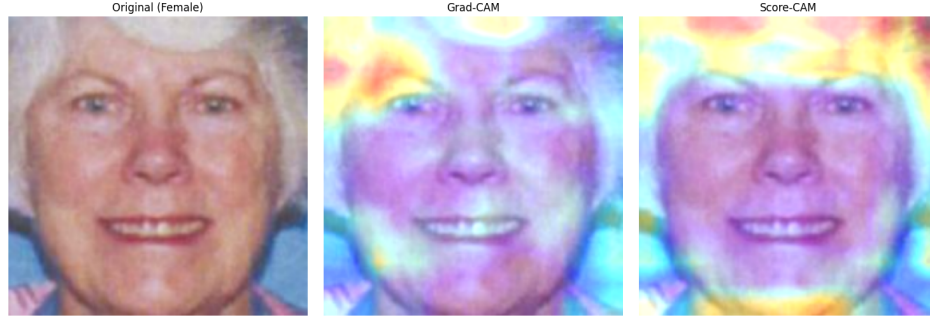


Figure 3.2: **Qualitative Comparison of CAM-based Explanation Methods.**

The analysis demonstrated that gender-based attention patterns stayed constant throughout the study, while age variables influenced on these patterns. The group-averaged heatmaps in Figure 3.3 demonstrate that the model focused its attention on lower and central facial regions when processing male faces. MorphoXAI detected peak activation points throughout the chin, jawline, and nose regions in all age groups, while its attention went primarily to the forehead, eyes and cheeks of female faces during processing. The "Teen" and "Middle-aged" age groups displayed the strongest gender-based attention patterns.

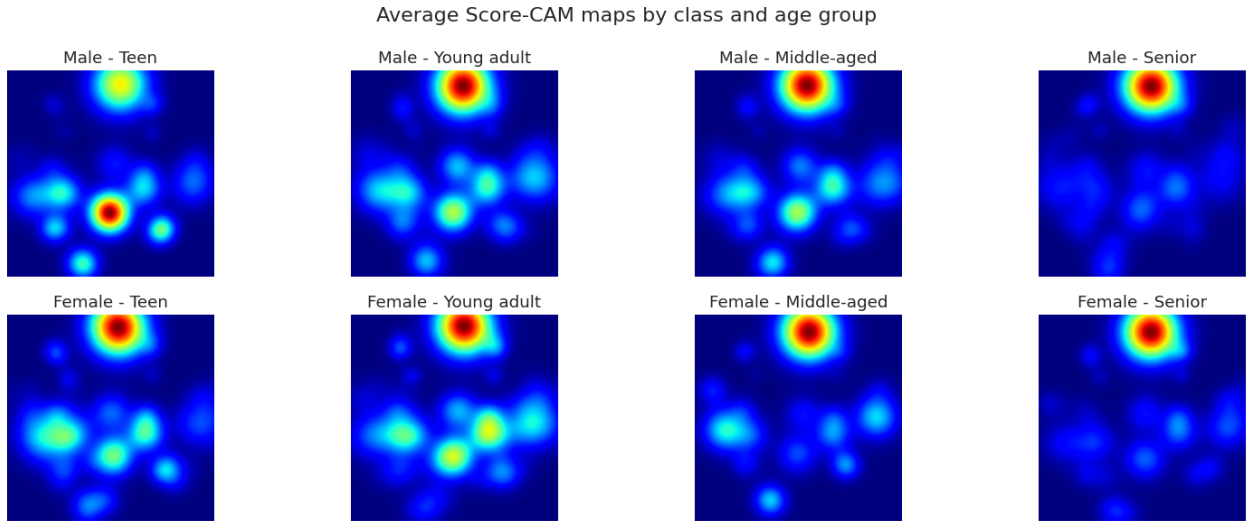


Figure 3.3: **Semantic Activation Scores by Gender and Age Group.**

3.1.3 Statistical Validation of Attentional Patterns

To statistically validate the observed attentional differences in the maps, a series of tests was executed on the 18-dimensional semantic activation vectors. The objective of this analysis is to go beyond visual inspection and confirm numerically the significance and predictive power of the anatomically defined features learned by the model.

First, a non-parametric Mann-Whitney U test was performed to confirm that the distributions of activation scores differed significantly between genders. Focusing on the results from Score-CAM due to its higher fidelity, the analysis found statistically significant differences in key dimorphic regions, most notably the chin centre ($p=0.0026$) and the right eyebrow ($p=0.0215$).

Also, a multivariate logistic regression analysis was implemented to determine the independent predictive power of each anatomical region for classifying a face as male, while controlling for all other regions. The results identified the right cheekbone as the strongest independent predictor, increasing the odds of a male classification by a factor of 12.15 (95% CI: [1.53 - 96.6]) for each standard deviation increase in attention.

Finally, a two-way ANOVA analysis showed a significant interaction effect between gender and age group across all 18 anatomical regions ($p<0.001$), indicating that the model’s application of gender features is regulated by the subject’s age.

3.1.4 Robustness and Latent Structure of Semantic Vectors

An important requirement for any XAI framework to be considered trustworthy is the stability of its explanations when it goes under input disturbances. In fact, the generated semantic vectors need to demonstrate a structured latent space organisation that aligns with natural data categories to be considered meaningful.

First, to determine if the model’s focus was functionally linked to its performance, the mean regional activations for correct and incorrect predictions were compared (Figure 3.4). The results show that for Score-CAM, successful classifications were associated with increased attention to key dimorphic regions such as the `right_jaw` and `right_eyebrow`. Conversely, a disproportionate focus on non-anatomical or background features (other category) was the strongest correlate of prediction errors, confirming that the model relies on relevant

anatomical features for reliable classification.

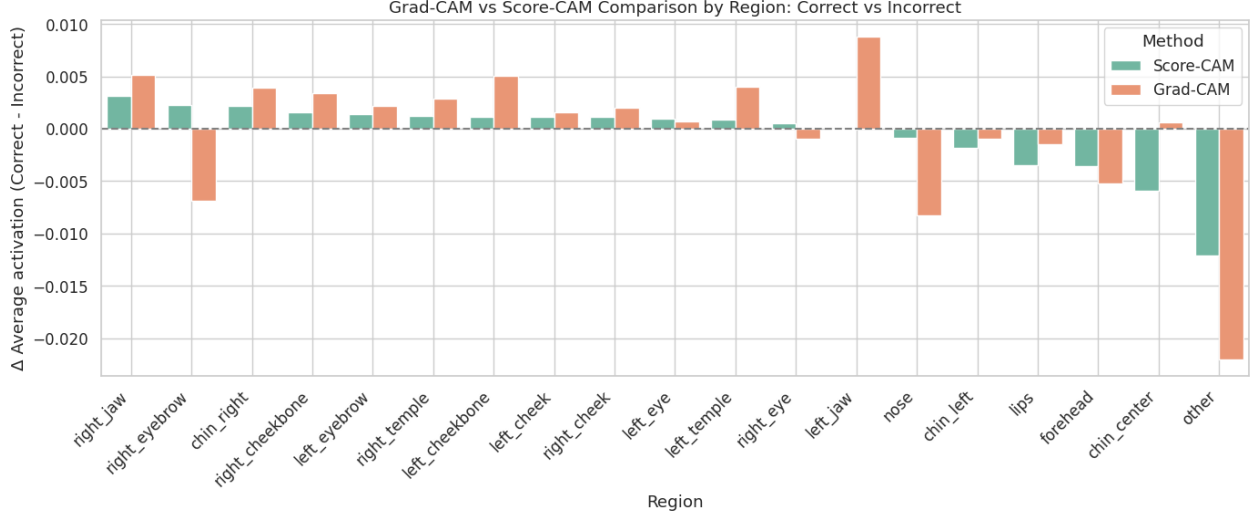


Figure 3.4: **Analysis of Model Attention for Correct vs. Incorrect Predictions.**

Second, the stability of the explanations was assessed under input deterioration via a Gaussian blur perturbation test. The framework’s robustness was outstanding when using Score-CAM, which maintained a median Pearson’s correlation of nearly 1.0 between the semantic vectors of original and blurred images. In contrast, Grad-CAM explanation showed a very wide distribution falling to correlations below 0.4. This finding confirms that the attention patterns captured by Score-CAM are a resilient signal, not an artefact of high-frequency image details.

Finally, dimensionality reduction techniques were applied to explore the latent structure of the semantic vectors. Contrary to initial expectations, the 2D projections from PCA and t-SNE analysis failed to show distinct gender clusters because the classes remained intermingled throughout the reduced dimensions (Figure 3.5). This indicates that while the semantic vectors contain significant predictive information, demonstrated in the statistical analysis, the decision ranges learned by the network is complex and high-dimensional, and cannot be captured by a simple linear or non-linear projection into two dimensions.

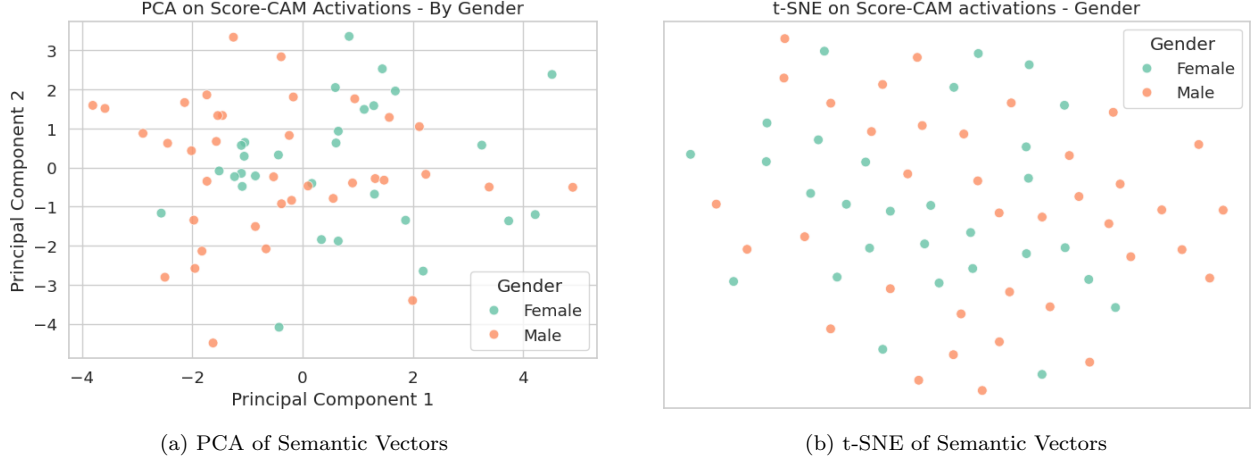


Figure 3.5: Embedding of Semantic Activation Vectors

3.2 Phase 2: Clinical Application on DBS Motor Recovery

3.2.1 Predictive Model Performance

The evaluation of 20 configurations between models and resolutions showed that classification performance was significantly correlated to image resolution, with the 224x224 pixel input size consistently resulting in the highest accuracies across all five architectures. The three models which achieved the highest results were Mobile-Net with 79.26% accuracy, ResNet50 with 82.97% and DenseNet-121 with 83.28% when using the optimal 224x224 pixel input size. These models significantly outperformed their lower-resolution counterparts and other architectures. Given its robust performance and architectural design that promotes feature reuse, DenseNet-121 at 224x224 resolution was selected as the champion model for a subsequent, in-depth analysis of its performance, calibration and explainability.

The selected DenseNet-121 model demonstrated a significant level of accuracy and reliability on the unseen, patient-disjoint test set. The model achieved 82.27% (95% CI: 78.72% - 85.8%) test set accuracy by using the optimal decision threshold of $\tau^*=0.79$, which was identified from the validation set. The confusion matrix analysis (Figure 3.6, Left) shows

a balanced performance, with the model correctly identifying 208 of 238 Pre-DBS images (True Negative Rate or Specificity of 87.4%) and 140 of 185 Post-DBS images (True Positive Rate or Sensitivity of 75.7%). The Precision-Recall (PR) curve analysis demonstrates the model’s strong discriminative power through its Area Under the Curve (AUC-PR) or Average Precision (AP) value of 88.02% (95% CI: 83.84% - 91.45%) as illustrated in Figure 3.6, Right.

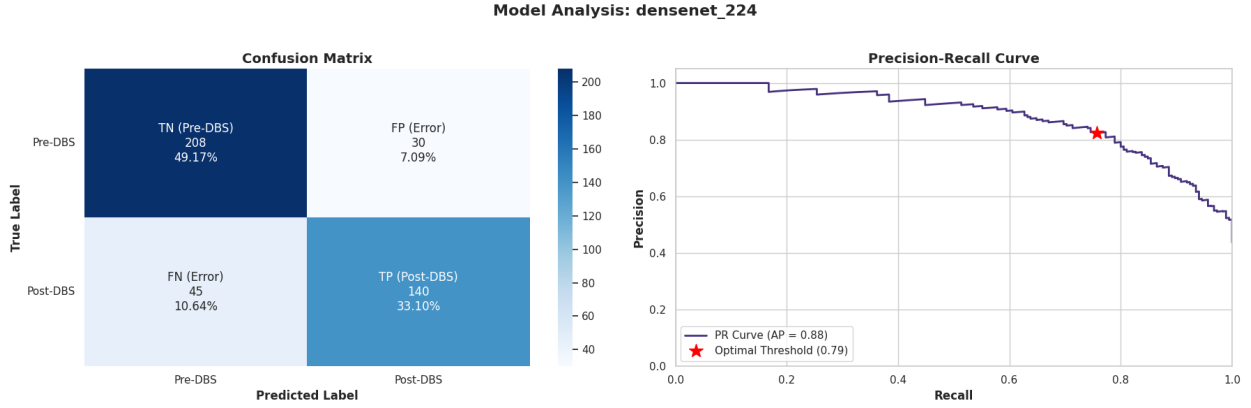


Figure 3.6: **Best model performance in DBS Classification.**

Beyond predictive accuracy, the reliability diagram (Figure 3.7) shows a reasonable correspondence between the model’s predictive confidence and its empirical accuracy. However, the statistical metrics showed a pattern of over-confidence, which is a common trait in modern neural networks. The model demonstrated this through an Expected Calibration Error (ECE) of 0.166 and a Brier Score of 0.167. The results show why post-hoc calibration techniques (such as Temperature Scaling) need to be applied before using these models in clinical practice, these results are presented in Table A.2.

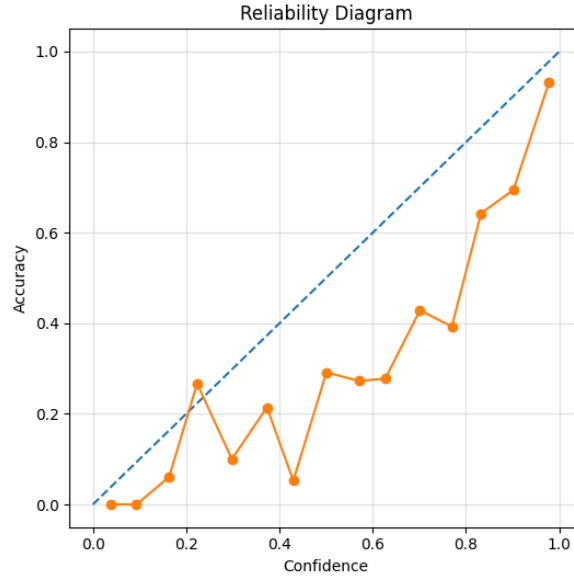
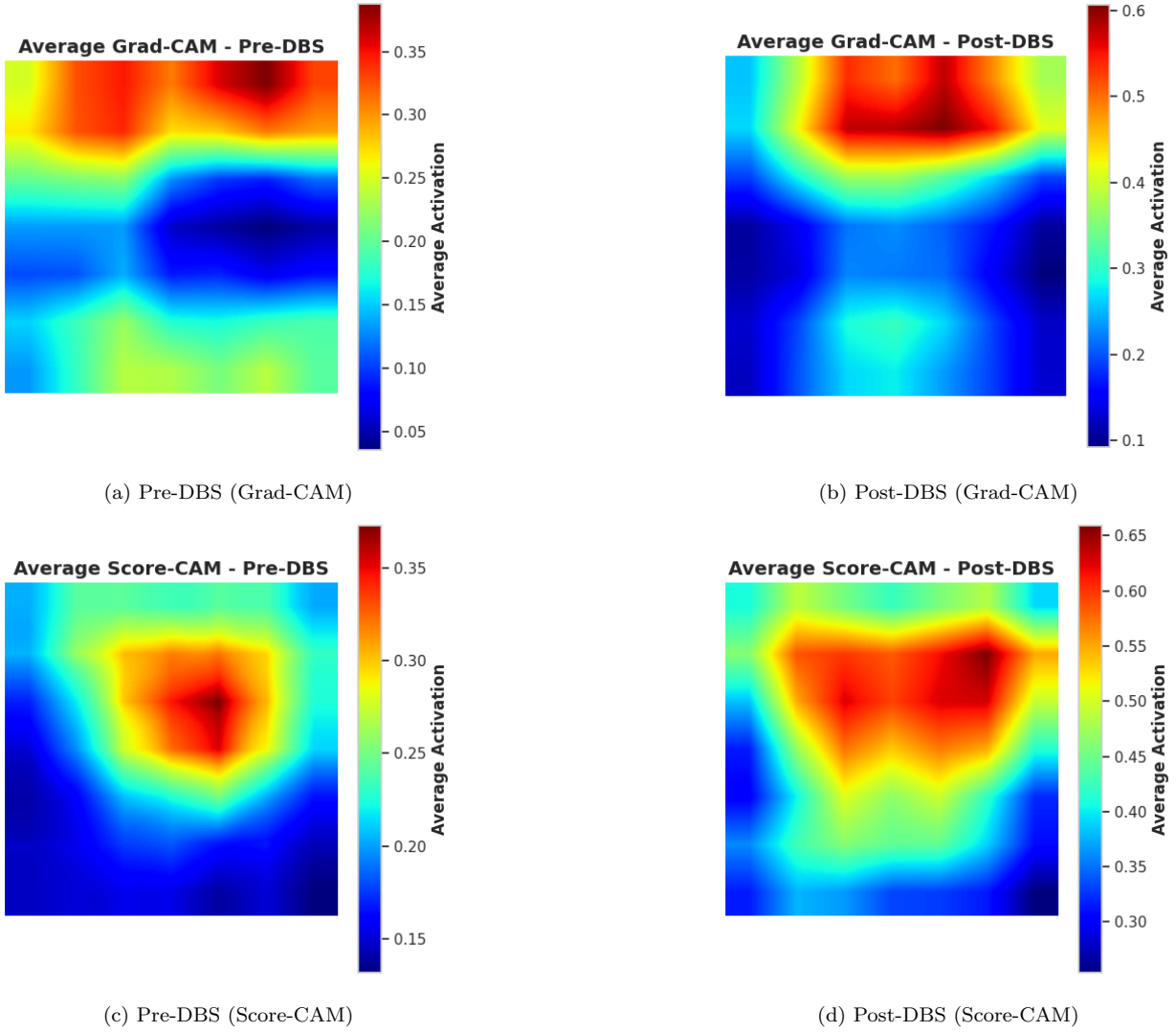


Figure 3.7: Reliability diagram of the DenseNet-121 Model.

3.2.2 Anatomical Attention Patterns Revealed by MorphoXAI

MorphoXAI application revealed a clinically important and methodical shift in attentional focus when applied to the best model's predictions. The model developed separate visual approaches for each clinical state according to the aggregated Class Activation Maps (Figure 3.8).

- For Pre-DBS images, the model's attention was focused on the central and upper facial regions, particularly the forehead and nasal bridge. This suggests the model may be recognising rigidity and lack of expression in these areas.
- For Post-DBS images, the model's focus shifted to the lower, expressive facial regions. Specifically, attention was reorganised towards the perioral (mouth and lips), zygomatic (cheeks), and mandibular (jawline) areas. These are the regions governed by the facial musculature critical for expression, such as smiling and speech articulation.

Figure 3.8: **Aggregated Class Activation Maps.**

The difference maps (Figure 3.9) give a detailed visualisation of this shift, where the intense red overlay on the lower face indicates a significant gain in feature importance for classifying the post-treatment state. This learned attentional shift provides strong evidence that the model is not depending on false correlations but has identified a valid, data-driven biomarker of facial motor recovery following DBS therapy.

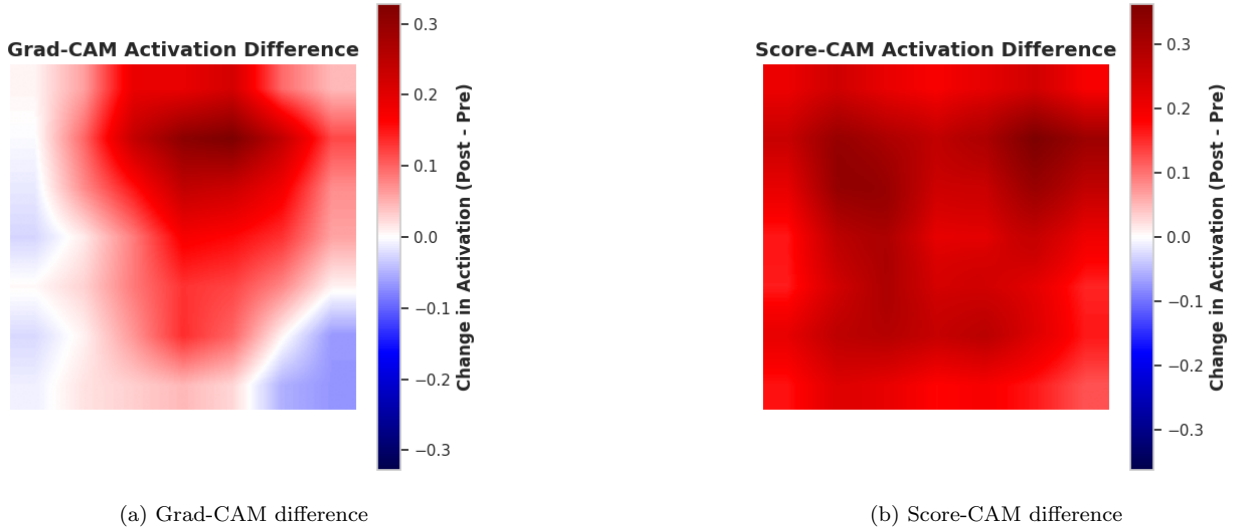


Figure 3.9: **Difference Maps Highlighting the Net Shift in Attention Post-DBS.**

3.2.3 Statistical Validation of Attentional Patterns

To statistically validate the changes observed in the aggregated heatmaps, a two-sided Mann-Whitney U test with Benjamini-Hochberg FDR correction was applied to compare the activation distributions of each of the 18 facial regions between the Pre- and Post-DBS groups. The analysis showed that XAI methods produced significant differences across the board, but they presented different patterns of sensitivity.

For Score-CAM, a total of 18 out of 18 regions have statistical significance ($p_{adj} < 0.05$). The results were controlled by regions in the left hemiface and lower-central areas, with the top three most significant regions being the left cheek ($p_{adj} = 3.96 \times 10^{-28}$), left cheekbone ($p_{adj} = 8.78 \times 10^{-26}$), and chin left ($p_{adj} = 2.09 \times 10^{-25}$).

For Grad-CAM, a smaller number of regions (10 out of 18) were found to be statistically significant. The method showed high sensitivity to the right hemiface and periocular regions. The top three most significant regions were the right eye ($p_{adj} = 6.16 \times 10^{-26}$), right eyebrow ($p_{adj} = 2.71 \times 10^{-20}$), and right cheekbone ($p_{adj} = 4.32 \times 10^{-13}$).

The results shows that Score-CAM detects more significantly left-side face changes while Grad-CAM detects right-side face changes.

To formally test if the effect of DBS on model attention was dependent on the specific

facial region, a two-way ANOVA was performed. The results confirmed a highly significant Group x Region interaction effect for both Score-CAM ($F(17, 7272) = 7.87, p < 10^{-20}$) and Grad-CAM ($F(17, 7272) = 12.08, p < 10^{-34}$). This statistical finding is crucial, as it gives definitive evidence against a uniform, global change in attention, proving instead that the model's learned strategy involves a complex and topographically specific redistribution of focus across the face as a function of the patient's clinical state.

3.2.4 Advanced Predictive Modelling and Sensitivity Analysis

To move beyond univariate comparisons and assess the independent predictive value of each anatomical region while accounting for multicollinearity, a multivariate modelling was employed. The primary analysis utilised an L2-penalised logistic regression model, with confidence intervals estimated via a patient-level bootstrap procedure.

The results, visualised in the forest plots (Figure 3.10), showed an important consistency across both XAI methods in identifying the right periocular area as the most salient biomarker of the post-DBS state. For Score-CAM, the right_eyebrow emerged as the strongest positive predictor ($OR = 2.93$). For Grad-CAM, the right_eye was the most significant ($OR = 2.67$). The models also identified significant negative predictors associated with the pre-DBS state, but with some variation: for Score-CAM was the nose, while for Grad-CAM, it was the left_temple.

To ensure these findings were not biased by the clustered nature of the data, a sensitivity analysis was conducted using a Generalised Estimating Equations (GEE) model. The GEE results were highly consistent with the model. The right_eyebrow (Score-CAM) remained strong and statistically significant ($p=0.008$). Notably, while showing a consistent effect size, the right_eye (Grad-CAM) was only slightly significant in the GEE model ($p=0.07$). This high level of concordance, even with minor variations, confirms the robustness of the finding that the right periocular region is an important biomarker for classifying the post-DBS state.

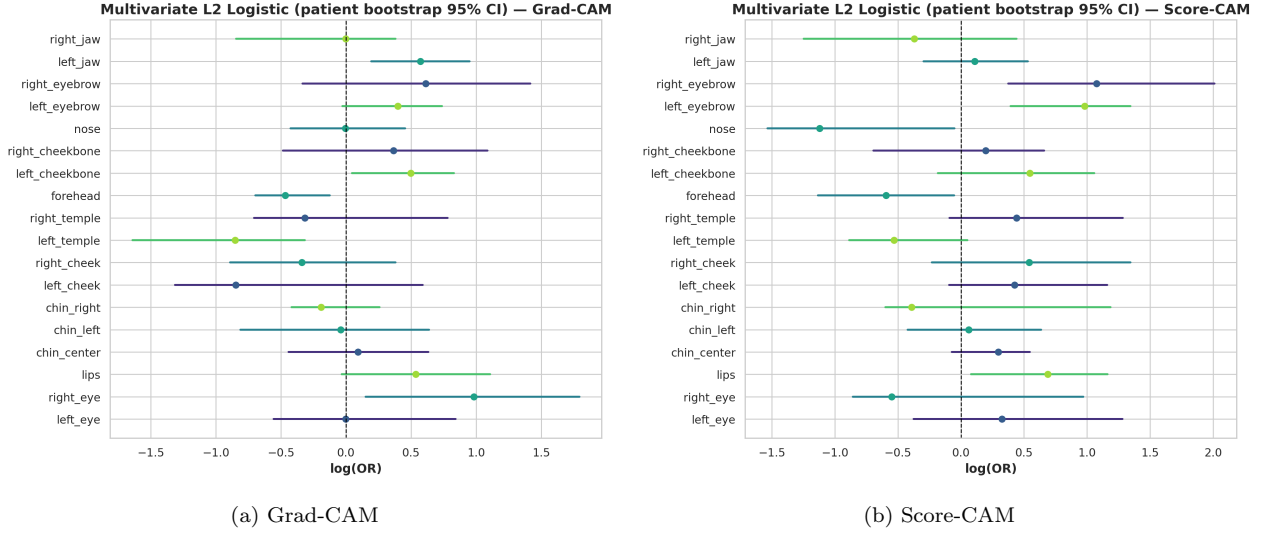


Figure 3.10: Multivariate L2 Logistic Regression Forest Plot

3.2.5 Robustness and Latent Structure of Semantic Vectors

The stability of generated explanations was demonstrated through a perturbation analysis using Gaussian blur. But first to assess that the model focus was linked to its performance the mean regional activations for correct and incorrect predictions were compared (Figure 3.11). The results for Grad-CAM showed a strong and clinically intuitive signal: correct predictions were associated with significantly higher activation in the left cheek (mean diff. = 0.0063), left jaw (mean diff. = 0.0047), and lips (mean diff. = 0.0033). This demonstrates that Grad-CAM’s success is tied to its focus on the expressive musculature. The findings for Score-CAM were more subtle, with very small activation differences, but still pointed towards the right eye and chin right as being associated with correct predictions. This indicates that while both methods present clinically relevant features, Grad-CAM’s decision process is more transparently linked to the primary areas of motor recovery.

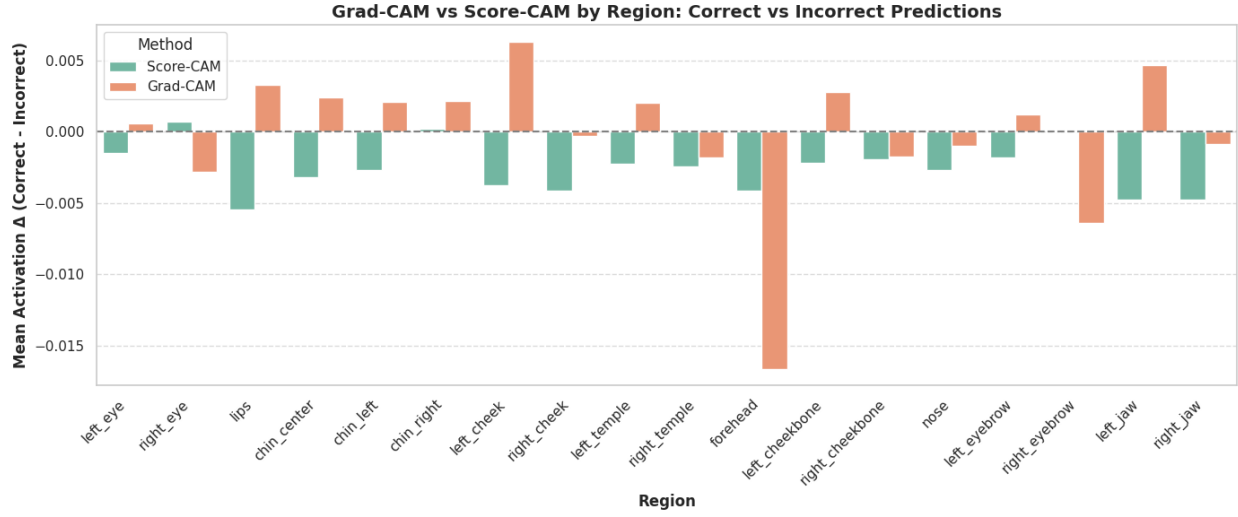


Figure 3.11: Analysis of Model Attention for Correct vs. Incorrect Predictions.

In Addition, the statistical analysis showed after the blur perturbation analysis that Grad-CAM generated more robust explanations because its Pearson’s correlation between original and blurred activations reached 0.615, whereas Score-CAM reached 0.478.

To analyse the underlying structure of the 18-dimensional semantic vectors, a dimensionality reduction was implemented. PCA showed that the top two principal components could explain a significant amount of the variance (Score-CAM: 55.9%; Grad-CAM: 48.6%) and showed a moderate separation between the clinical groups (Figure 3.12). The non-linear technique t-SNE showed this separation with much greater clarity. The t-SNE embedding showed that Score-CAM and Grad-CAM learned attentional patterns creating distinct clusters that are directly in concordance with the clinical state (Figure 3.13). This proves that the semantic vectors are not random noise but contain a rich, discriminative structure.

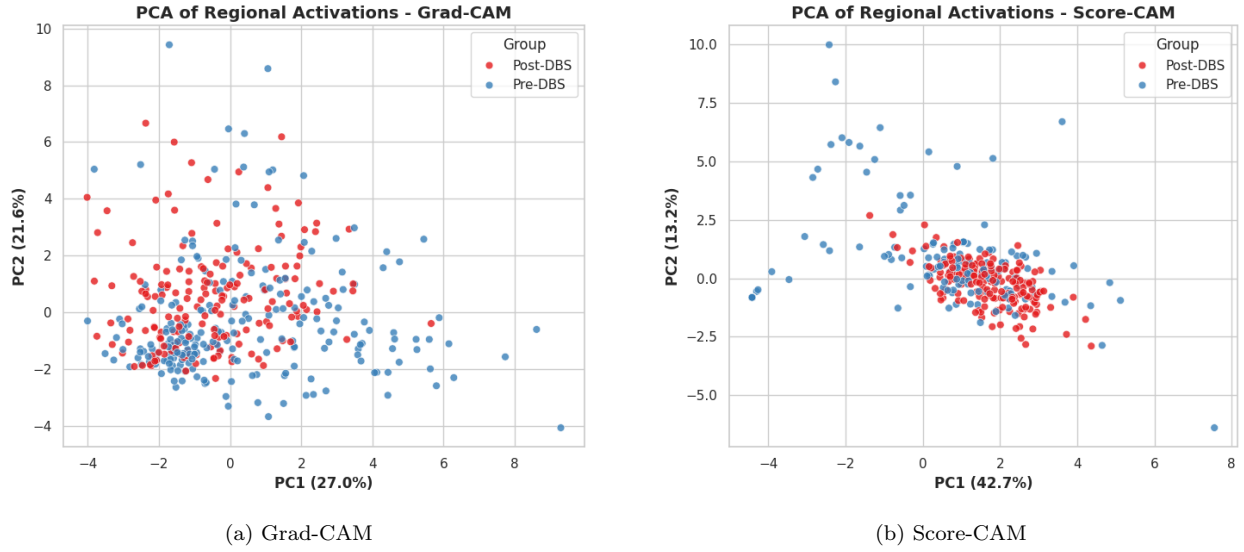


Figure 3.12: PCA of Regional Semantic Activation Vectors

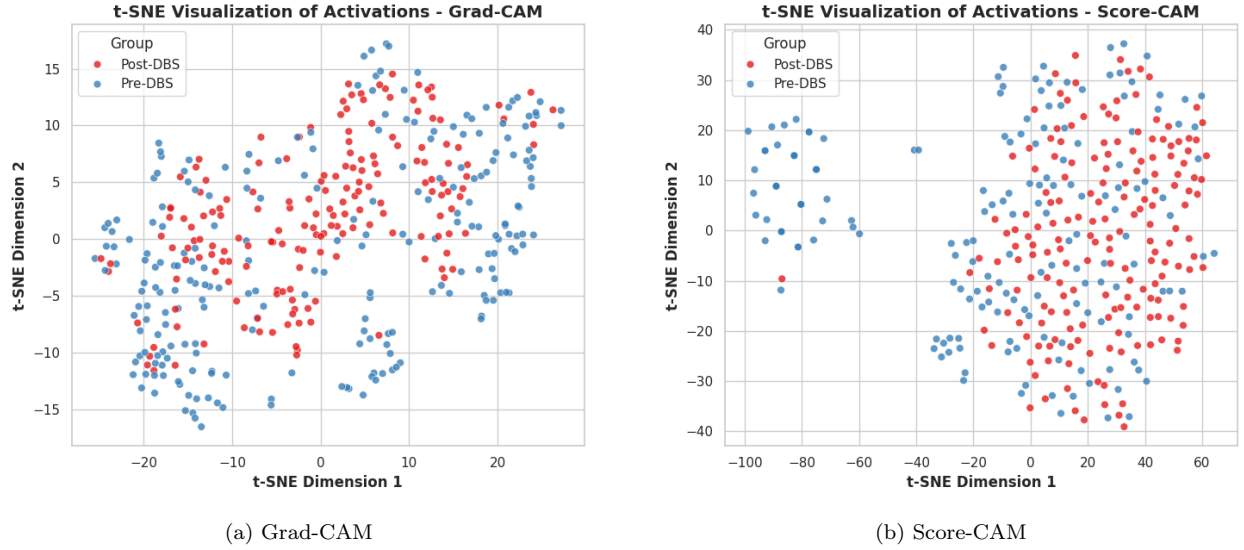


Figure 3.13: t-SNE Visualisation of Learned Attentional Representations.

Unlike Phase 1, where Score-CAM resulted in higher stability, Grad-CAM presented a stronger correlation during the DBS phase, indicating that the stability depends on the specific task and domain and should be empirically verified rather than assumed.

3.2.6 Unsupervised Discovery of Attentional Archetypes

Finally, unsupervised K-Means clustering was applied to the semantic vectors to identify the continuous "attentional archetypes" (Figure 3.14). For Score-CAM, the analysis discovered two distinct Pre-DBS patterns and one primary Post-DBS pattern. Clusters 1 and 2 were almost exclusively composed of Pre-DBS cases (99% and 95% respectively). In contrast, Cluster 0 was clearly the "recovered expression" pattern, with 62% of its members being Post-DBS cases. A similar, though less sharply defined, correspondence was found for Grad-CAM. This last result demonstrates that MorphoXAI not only produces robust explanations but also discovers consistent and clinically meaningful explanatory strategies employed by the model, which are discoverable through unsupervised methods.

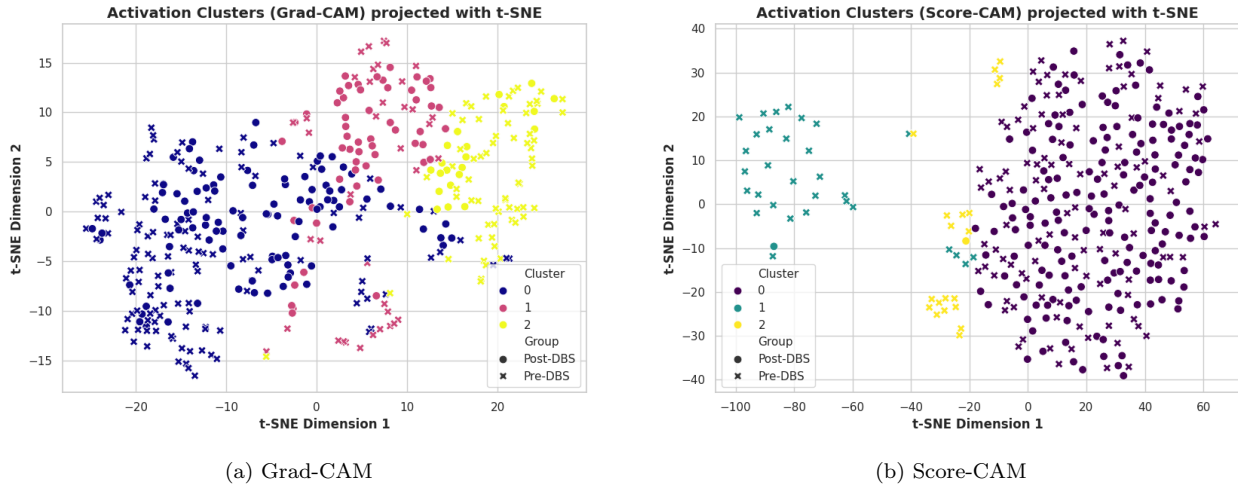


Figure 3.14: Unsupervised Clustering.

Chapter 4

Discussion

4.1 Synthesis and positioning with respect to the literature

The finding shows that converting pixel-level attributions into *anatomy-aligned semantic vectors* produces a stable medical representation of model evidence that corresponds to craniofacial regions. This representation directly addresses documented weaknesses of pixel-wise saliency in medical imaging—limited faithfulness, instability, and misalignment with clinicians’ conceptual vocabulary (Adebayo, J. Gilmer, and Muelly, 2018; Hooker and Erhan, 2019; Kindermans, 2019). Unlike qualitative overlays, region scores enable hypothesis testing, error signature analysis, and comparisons across operating thresholds, delivering relevant decision transparency that has been largely unknown from other facial phenotyping systems. The development of morphology-aware interpretability represents a logical progression from previous automated facial diagnosis systems (e.g., DeepGestalt and subsequent prospective assessments) on the way to building *trustworthy* clinical AI systems (Gurovich et al., 2019; Ferry et al., 2022; Rajpurkar et al., 2022; Kelly et al., 2019).

4.2 Interpretation of principal findings

The process of aggregating attribution within predefined ROIs during different processes detected standard morphological features that are used in dysmorphology and clinical

facial analysis on a daily basis, such as nasal bridge and mandibular contour (Pantelis et al., 2021). In *Phase 1 (sex classification)*, semantic vectors showed continuous emphasis on mandibular, mental (chin), and nasal-bridge regions for male faces, contrasted with periocular/forehead and zygomatic patterns for female faces—trends consistent with facial-attribute and recognition literature that highlights region-specific cues and shape differentials (Pantelis et al., 2021; S. e. a. Liu, 2021; M. Wang and W. Deng, 2021). In *Phase 2 (clinical)*, a post-intervention redistribution of decision attention regarding lower-face ROIs (perioral/zygomatic/mandibular) was detected, consistent with improvement in *hypomimia* (facial masking) and the recovery of expressive musculature reported in Parkinson’s disease research (Scherpf, 2019; Abreu, 2021; Tolosana, 2022). These results, quantified at the ROI level, provide a physiologically plausible reading that would be difficult to extract from raw heatmaps alone and support the use of anatomy-grounded vectors as an auditable intermediate layer between black-box predictions and clinician-facing interpretation.

4.3 Faithfulness, Stability, and Positioning MorphoXAI

Complementing the literature synthesis in §4.1, this section assesses how MorphoXAI behaves in practice and why it is operationally preferable for clinical translation, grounding all claims in our experiments and in the projection/statistical protocol defined in Methods (Secs. 2.1.2 and 2.4.1). Evidence is drawn from Phase 1 (Face2Gender; Secs. 3.1.2–3.1.3) and Phase 2 (DBS; Secs. 3.2.2–3.2.3). Calibration aspects are treated in §4.4 and implementation details in §2.3.5.

Regarding **faithfulness**, we rely on behavioural probes rather than subjective inspection. ROI-guided deletion and insertion consistently showed that suppressing regions identified as important by the semantic attribution vector degraded predictions more than suppressing low-importance regions, while reintroducing high-importance regions restored a larger fraction of the original score. In Face2Gender (Secs. 3.1.2–3.1.3), this monotonic effect held across operating points, indicating that the regional scores capture evidence the classifier actually uses. In DBS (Secs. 3.2.2–3.2.3), the most pronounced probability drops concentrated in peri-oral and peri-ocular areas, aligning with clinically expected changes in hypomimia. Because the semantic vector is tied to a fixed facial ontology (Sec. 2.1.2), cohort aggregates (means and confidence intervals) formed stable, clinically legible narratives, and

ROI-wise contrasts were tested with multiple-comparison control as specified in Sec. 2.4.1. Uncertainty was reported alongside discrimination in Sec. 3.2.1, moving explanations from anecdotal overlays to statistically auditable summaries.

Concerning **stability**, explanations remained robust under benign sources of variability. Across independent random initialisations, semantic vectors exhibited high similarity, particularly in the large-scale, in-the-wild setting of Face2Gender where Score-CAM produced more focal regional signals and higher across-seed similarity (Sec. 3.1.3). Under patient-disjoint partitions (Sec. 2.3.1), ROI aggregates and their confidence intervals overlapped across validation and test sets; in DBS (Sec. 3.2.2), peri-oral and peri-ocular emphasis persisted across splits, indicating that our summaries are not artefacts of a single partition. Because projection quality depends on face parsing, we stress-tested segmentation with mild perturbations and observed that cohort-level confidence bands absorbed residual variance, supporting the recommendation to institute routine QC for segmentation and to preserve a stable ROI taxonomy across studies (Sec. 2.1.2). Finally, the choice of pixel-level explainer is itself part of the design; our results show context dependence—Score-CAM is more stable in Face2Gender, whereas Grad-CAM correlates better in DBS under the same audit (Secs. 3.1.3 and 3.2.3)—aligning with calls to validate attribution choices empirically rather than by convention (Adebayo, J. Gilmer, and Muelly, 2018; Hooker and Erhan, 2019).

For **clinical positioning**, we focus on what our pipeline enables operationally. Because MorphoXAI projects attributions onto a fixed facial ontology (Sec. 2.1.2), each dimension of the semantic vector has a stable anatomical meaning across patients and training runs. This makes routine governance feasible: aggregation and uncertainty quantification at ROI level; multiple-testing control for cohort contrasts (Sec. 2.4.1); stratification by calibrated operating thresholds (see §4.4); and longitudinal monitoring of attribution drift. In practice, this translates into clinician-facing summaries that can be audited, compared across sites or time, and linked to decision policies, rather than one-off images that are difficult to quantify. Importantly, these benefits come without the heavy annotation burden of concept-bottleneck approaches and with clearer cohort quantification than prototype/case-based explanations, while remaining compatible with legacy datasets and small clinical cohorts.

The **context and threats to validity** are explicit. Task and domain shape explainer behaviour: Score-CAM proved advantageous in a large, heterogeneous benchmark, while Grad-CAM aligned better with the subtler DBS signal (Secs. 3.1.3 and 3.2.3). The quality of

the projection depends on the detection/landmarking process; the mitigations include robust parsers, rejection of low-confidence masks, and reporting cohort-level confidence intervals to absorb residual variance (Sec. 2.1.2). Because explanations are confined to the predefined ROI ontology, extra-facial or micro-textural cues may be under-represented; accordingly, we recommend dual reporting—ROI vectors as the primary, audit-ready layer plus a lightweight saliency sanity-check—and hypothesis-driven concept tests where appropriate.

In summary, MorphoXAI yields anatomy-aligned, auditable signals that can be statistically tested, calibrated, and monitored over time. Its limitations are explicit and manageable within standard validation practice, and its strengths—quantitative auditability, anatomical grounding, and low annotation burden establish it as a practical default explainer for facial phenotyping tasks.

4.4 Calibration, operating thresholds, and decision relevance

The reliability of clinical assessments depends in the same proportion on calibration and discrimination performance (Murphy et al., 2021; Chan, 2020; Y.-L. e. a. Chen, 2023). Because explanation patterns vary with the operating point, region-level analyses were conducted at *calibration-selected* thresholds, after inspecting reliability diagrams and summary calibration errors. The sensitivity of ROI scores to threshold shifts demonstrated that explanatory states need to be linked to *clinically justified operating thresholds* (e.g., precision–recall trade-offs suitable for triage) and validated on held-out data rather than on random cuts (Rajpurkar et al., 2022).

4.5 Generalisation, dataset shift, and fairness

Medical vision faces a major performance risk caused commonly by *dataset shifts* and *demographic imbalance* (Geirhos, 2020; D’Amour, 2020; Taori, 2020; Koh, 2021; Hendrycks and Dietterich, 2019). In the same way, facial models present extreme sensitivity to population composition and acquisition conditions, which provokes documented performance differences based on sex and ancestry (Buolamwini and Gebru, 2018; Larrazabal, 2020; Cirillo,

2020; Mehrabi, Morstatter, and Saxena, 2021). That’s why ROI vectors offer two advantages for addressing these concerns:

1. **Worst-group diagnostics:** Evidence from different regions allows for demographic strata analysis (e.g., sex, age groups, ancestry proxies) to detect failure modes that aggregate metrics hide.
2. **Mechanistic audits:** Domain-specific confounders such as background, pose, and illumination become identifiable through tracking the changes in which regions carry decision weights, which leads to focused solutions like background suppression, alignment and region-specific augmentation.

Robustness interventions—including *MixUp*, *CutMix*, and *AugMix*—should be evaluated for their effect on both accuracy and the *stability* of ROI evidence, as well as on *majority–minority gaps* (H. Zhang et al., 2017; Yun, Han, and Oh, 2019; Hendrycks and Mu, 2020). In cohorts with pronounced group shifts, distributionally robust training can complement these audits by improving worst-case generalisation (Sagawa, 2019). Throughout, transparent documentation—*model cards*, *data sheets*, and related governance artefacts—should accompany results to clarify intended use, limitations, and fairness evaluations (Mitchell, 2019; Gebru and Morgenstern, 2021; Arnold, 2019).

4.6 Clinical implications and intended use

Model evidence expressed through anatomical terms enhances the interpretation and communication of results, which allows reviewers to examine some failure settings, like false positives and negatives, in particular ROIs. In *genetic screening* tasks, calibrated probabilities along with ROI vectors can help *triage* (e.g., test prioritisation, referral decisions) provided that the scope of use, population coverage, and uncertainty are made explicit (Kelly et al., 2019; Price, Gerke, and Cohen, 2022). In *Parkinson’s disease*, the observed post-intervention shift toward lower-face expressive muscles offers hypothesis-generating insight into the recovery of facial dynamics; however, translation to clinical decision support would need expected validation with standardised endpoints and controls for medication state and comorbidities (Scherpf, 2019; Abreu, 2021; Tolosana, 2022). In all cases, the outputs need to include documentation that follows governance frameworks

to enable auditing of limitations and provenance and known failure modes (Mitchell, 2019; Gebru and Morgenstern, 2021; Arnold, 2019).

4.7 Limitations

Several limitations across data, methodology, evaluation, and implementation restrict the interpretability and external validity of the reported findings.

First, **data-level validity** is bounded by dataset composition. Generalisation is reduced by limited demographic diversity in the training process, which restricts the strength of fairness conclusions and increases the risk of performance disparities across underrepresented groups (Mehrabi, Morstatter, Saxena, et al., 2021; Larrazabal et al., 2020). Furthermore, heterogeneity in image acquisition conditions (e.g., illumination, pose) introduces potential confounders that models may use as shortcut signals unrelated to clinical morphology. The absence of external, multi-centre validation means that vulnerability to dataset shift remains a credible threat to deployment beyond the studied domains (Geirhos, 2020; Taori, 2020).

Second, the **methodological approach** has natural limitations. The framework relies on post-hoc attribution, meaning the explanations are correlational rather than causal and inherit the biases of the base classifier. The projection from saliency to region-of-interest (ROI) vectors also presupposes robust face detection and landmarking, and its fidelity may decrease under challenging conditions such as occlusions or extreme poses. Moreover, the 2D anatomical atlas is an approximation of inherently 3D craniofacial structures, which may lead to residual mismatch.

Finally, the **Evaluation and implementation** stage introduces further considerations. The internal validity may be affected by unmeasured confounding variables, while external validity remains restricted because of the lack of independence caused by trials in multi-centre cohorts. Regarding the implementation, specifically facial images, are naturally identifiable, provoking a risk to privacy and governance that persists even after anonymisation. That is why deployment of responsible facial recognition systems requires secure processing pipelines along with strict access control measures and clear data retention policies to meet prevailing governance principles (Gong, S. Liu, and H. e. a. Wang, 2021).

4.8 Future work

Methodological extensions include integrating *concept-level* techniques (e.g., *TCAV*, *concept bottleneck models*, *regression concept vectors*) on top of ROI embeddings to bridge from regional evidence to clinical concepts (Kim, Wattenberg, and J. e. a. Gilmer, 2018; P. W. e. a. Chen, 2020; Graziani, 2020; Bau and Zhou, 2017; Ghorbani, 2019). Carefully constrained *counterfactual* analyses could probe the necessity/sufficiency of regions while avoiding unsafe generative manipulations. *Uncertainty quantification* (temperature scaling, data-driven calibration, shift detectors) should be coupled to interpretability so that regional evidence is flagged or suppressed under covariate shift (Chan, 2020; Y.-L. e. a. Chen, 2023; Koh, 2021). From a data perspective, *multicentre*, *multi-ethnic* cohorts and *prospective* studies are required to characterise external validity and subgroup behaviour; robust augmentation and domain generalisation should be assessed for both accuracy and explanation stability (Hendrycks and Dietterich, 2019; Taori, 2020; Oala, 2022). For deployment, governance artefacts should be expanded with routine bias audits, periodic recalibration, and post-market monitoring (Mitchell, 2019; Gebru and Morgenstern, 2021; Arnold, 2019; Price, Gerke, and Cohen, 2022). Finally, *Human-factors* evaluations with clinicians need to assess whether ROI explanations enhance task performance and reduce over-reliance and improve team calibration better than heatmap-only baselines (Rajpurkar et al., 2022).

Chapter 5

Conclusions

A central barrier to the clinical translation of automated facial phenotyping has been the opacity of high-performing deep models. This dissertation addressed that barrier by introducing **MorphoXAI**, a framework that projects post-hoc attributions into *anatomy-aligned semantic vectors* over predefined craniofacial regions of interest (ROIs). By quantifying where decision evidence concentrates in anatomical terms, explanations were rendered statistically auditable and clinically legible, moving beyond anecdotal pixel overlays toward *testable* evidence (Adebayo, J. Gilmer, and Muelly, 2018; Hooker and Erhan, 2019; Kindermans, 2019).

Empirical results across two complementary studies supported discriminative value and physiological plausibility. In a Face2Gender setting, ROI vectors emphasised mandibular/mental and nasal-bridge evidence for male faces and periocular/forehead and zygomatic evidence for female faces, echoing established dysmorphological descriptors (Pantelis et al., 2021). During the hypomimia scenario (Phase 2), decision evidence moved toward lower-face ROIs, including perioral/zygomatic/mandibular areas, which confirmed better facial expressivity and re-engagement of important muscles (Scherpf, 2019; Abreu, 2021; Tolosana, 2022). These trends, difficult to validate from raw heatmaps, became statistically tractable once projected into ROI space.

Reliability and decision relevance improved through the connection of interpretation to *calibrated* operating points and the auditing of explanation stability across different attribution families as well as preprocessing methods and resolution and feature layers which established interpretability as an empirical design choice that needs justification

(Murphy et al., 2021; Y.-L. e. a. Chen, 2023; Rajpurkar et al., 2022). At the same time, evidential strength remains bounded by dataset composition and post-hoc attribution; external validity across institutions and devices is required, and privacy risks inherent to facial imagery motivate secure pipelines and governance-aligned documentation of intended use and limitations (Gong, S. Liu, and H. e. a. Wang, 2021; Mitchell, 2019; Gebru and Morgenstern, 2021).

The trajectory is concrete: advancing from ROIs to *concept-level* alignment (e.g., TCAV or concept bottlenecks atop ROI embeddings), carefully constrained counterfactuals to probe necessity/sufficiency, uncertainty-aware explanations under shift, and prospective, multicentre evaluations (including video-based facial dynamics) alongside human-factors studies of clinician performance. In sum, *quantitative anatomy-aligned explanations* serve as a practical solution between current vision performance and medical transparency needs to develop accurate and auditable facial phenotyping systems that deliver equitable and clinically interpretable results (Rajpurkar et al., 2022; Mitchell, 2019; Gebru and Morgenstern, 2021).

Appendix A

A.1 Additional Figures

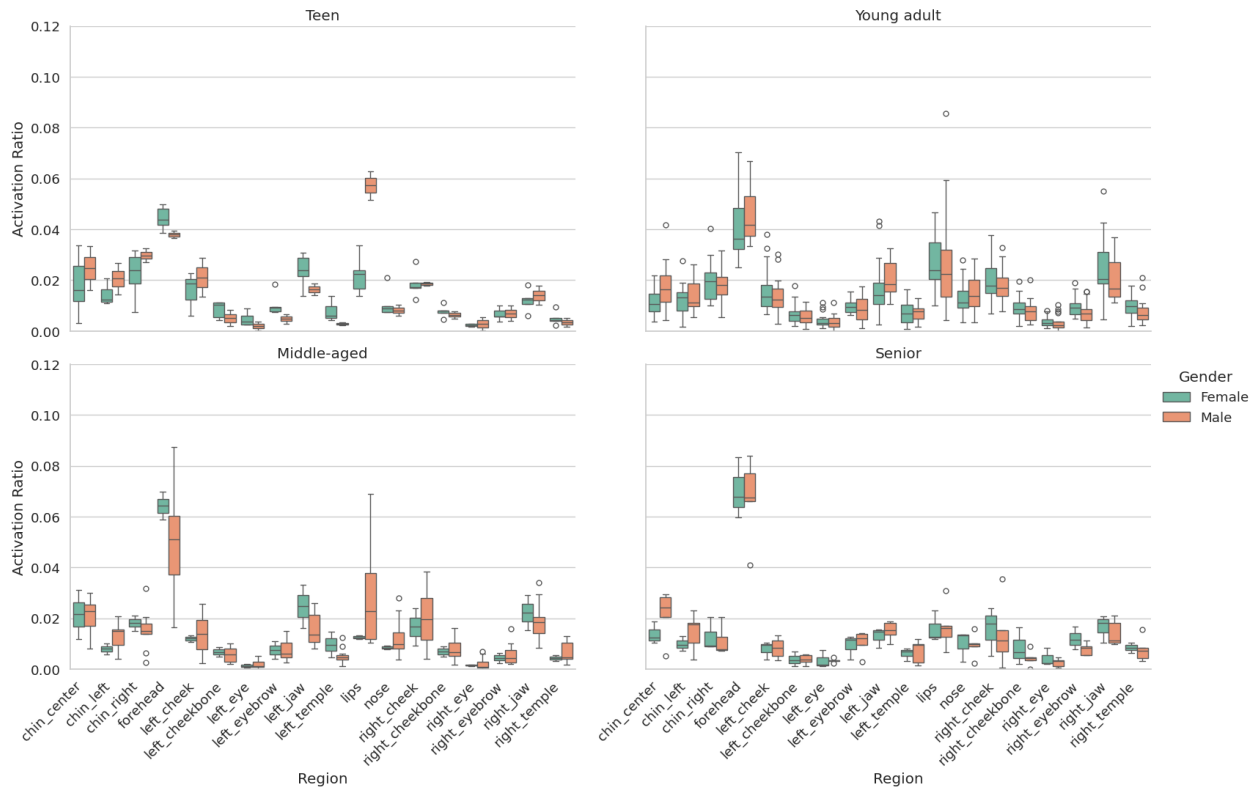


Figure A.1: Semantic Activation Scores by Gender and Age Group using Score-CAM (Phase 1)

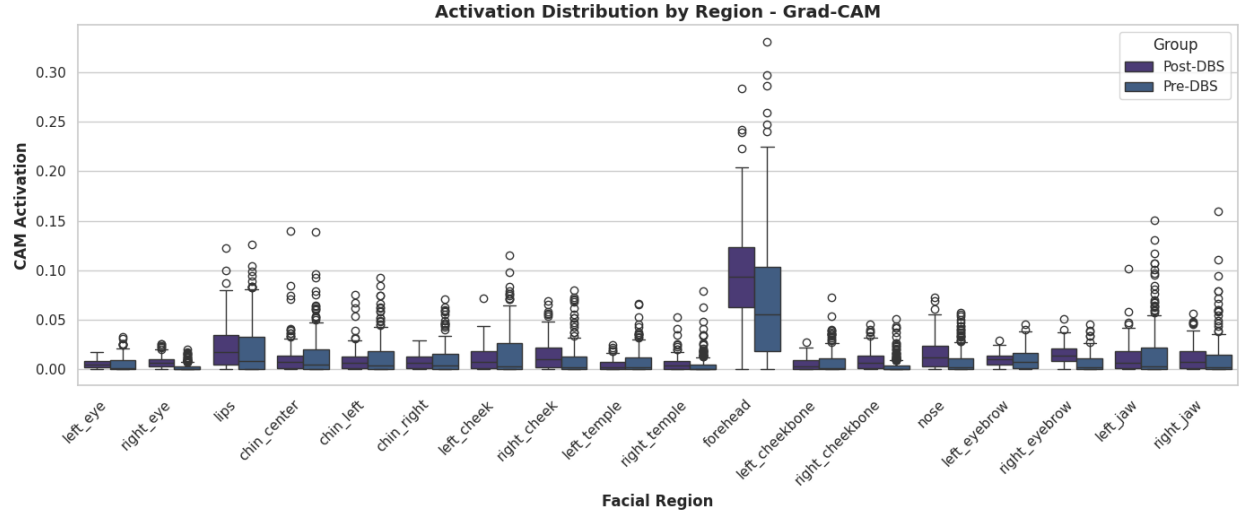


Figure A.2: Distribution of Regional Grad-CAM Activations by Clinical Group (Phase 2).

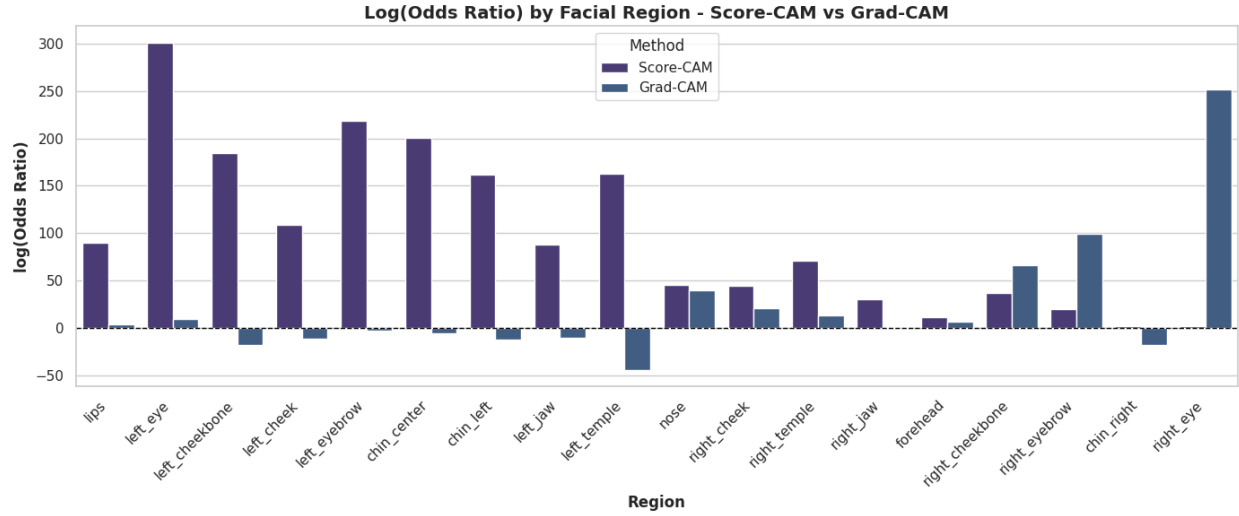


Figure A.3: Univariable Log(Odds Ratios) for Predicting Post-DBS Status (Phase 2).

A.2 Additional Tables

Region	Score-CAM		Grad-CAM	
	OR	p	OR	p
right_cheekbone	12.15 (1.53, 96.58)	0.02	1.10 (0.12, 9.71)	0.93
left_cheekbone	3.77 (0.69, 20.78)	0.13	226.29 (5.39, 9505.35)	0.00
right_eyebrow	2.56 (0.73, 8.87)	0.14	0.22 (0.02, 1.95)	0.30
left_temple	1.50 (0.43, 5.21)	0.47	1.10 (0.32, 3.76)	0.88
right_temple	1.50 (0.48, 4.52)	0.47	3.36 (0.48, 23.64)	0.22
left_cheek	1.01 (0.27, 3.73)	0.99	0.00 (0.00, 0.22)	0.01
chin_right	0.95 (0.26, 3.49)	0.93	5.67 (0.72, 44.43)	0.10
left_jaw	0.78 (0.22, 2.75)	0.30	29.24 (1.69, 505.39)	0.00
left_eye	0.52 (0.14, 1.67)	0.27	1.77 (0.22, 14.51)	0.60
right_cheek	0.45 (0.08, 2.44)	0.36	1.02 (0.12, 8.92)	0.97
lips	0.30 (0.06, 1.40)	0.13	2.20 (0.21, 23.25)	0.39
forehead	0.28 (0.08, 0.97)	0.04	3.05 (0.30, 30.51)	0.34
left_eyebrow	0.25 (0.07, 0.92)	0.04	4.32 (0.74, 17.67)	0.11
right_jaw	0.24 (0.06, 0.97)	0.05	1.53 (0.04, 55.79)	0.64
other	0.22 (0.03, 1.44)	0.11	2.04 (0.10, 43.82)	0.65
chin_left	0.16 (0.04, 0.67)	0.10	0.28 (0.06, 1.42)	0.45
nose	0.11 (0.03, 0.51)	0.00	1.48 (0.17, 12.17)	0.72
chin_center	0.10 (0.02, 0.48)	0.00	1.42 (0.28, 7.10)	0.76
right_eye	0.08 (0.01, 0.51)	0.00	0.40 (0.04, 3.35)	0.64

Table A.1: Odds ratios (OR) with 95% CIs for Score-CAM and Grad-CAM (Phase 1).

Metric	Value
val_acc	83.28%
threshold	0.79
test_acc	82.27% (78.72% - 85.82%)
AP	88.02% (83.84% - 91.45%)
ECE	0.17
Brier	0.17
T°	1.21

Table A.2: Performance and Calibration Metrics for the DenseNet-121 Model (Phase 2).

	sum_sq	df	F	PR(>F)
C(Group)	0.007989	1.0	2.133796	4.455761e-06
C(Region)	2.133796	17.0	331.350841	0.000000e+00
C(Group):C(Region)	0.077804	17.0	12.081971	7.993730e-34
Residual	2.754671	7272.0	NaN	NaN

Table A.3: Full Grad-CAM Two-Way ANOVA Results (Phase 2)

Region	P_value_grad	P_value_score	P_adj_grad	★	P_adj_score	★
left_cheek	1.217585e-01	2.199801e-29	1.565467e-01	-	3.959643e-28	✓
left_cheekbone	3.696801e-02	9.755302e-27	6.049311e-02	-	8.779772e-26	✓
chin_left	1.733169e-01	3.475795e-26	1.942834e-01	-	2.085477e-25	✓
chin_center	1.834899e-01	3.981028e-25	1.942834e-01	-	1.791462e-24	✓
left_eye	3.933863e-04	1.053304e-24	7.867726e-04	✓	3.791894e-24	✓
lips	6.573629e-03	1.459436e-24	1.183253e-02	✓	4.378309e-24	✓
left_eyebrow	6.821339e-02	2.109138e-23	1.023201e-01	-	5.423499e-23	✓
left_jaw	8.849067e-02	4.144786e-23	1.225255e-01	-	9.325768e-23	✓
chin_right	1.365868e-01	1.126475e-18	1.639042e-01	-	2.252951e-18	✓
left_temple	8.784013e-01	2.373936e-18	8.784013e-01	-	4.273085e-18	✓
right_cheek	7.552452e-07	7.888383e-16	1.942059e-06	✓	1.290826e-15	✓
right_temple	1.948250e-08	4.184397e-14	7.013700e-08	✓	6.276595e-14	✓
nose	1.155888e-10	5.478775e-14	5.201495e-10	✓	7.585996e-14	✓
right_cheekbone	7.208013e-14	3.550388e-12	4.324808e-13	✓	4.564784e-12	✓
right_jaw	3.215044e-04	1.357034e-10	7.233849e-04	✓	1.628441e-10	✓
right_eyebrow	3.011836e-21	1.143845e-09	2.710652e-20	✓	1.286826e-09	✓
right_eye	3.423194e-27	6.752905e-08	6.161749e-26	✓	7.150135e-08	✓
forehead	4.416299e-08	1.773632e-06	1.324890e-07	✓	1.773632e-06	✓

Table A.4: Statistically Significant Regions from Mann-Whitney U Test (Phase 2).

Region	L2-Penalised Model		GEE Model		
	OR	log(OR)	OR	log(OR)	p-value
left_eye	1.00 (0.57, 2.32)	-0.00 (-0.56, 0.84)	0.99 (0.50, 1.97)	-0.01 (-0.69, 0.68)	0.98
right_eye	2.67 (1.16, 6.01)	0.98 (0.15, 1.79)	2.72 (0.91, 8.18)	1.00 (-0.10, 2.10)	0.07
lips	1.71 (0.97, 3.03)	0.54 (-0.03, 1.11)	2.04 (0.89, 4.63)	0.71 (-0.11, 1.53)	0.09
chin_center	1.10 (0.64, 1.88)	0.09 (-0.44, 0.63)	1.07 (0.57, 2.01)	0.07 (-0.56, 0.70)	0.84
chin_left	0.96 (0.44, 1.89)	-0.04 (-0.81, 0.64)	0.97 (0.39, 2.41)	-0.03 (-0.94, 0.88)	0.94
chin_right	0.83 (0.66, 1.29)	-0.19 (-0.42, 0.26)	0.85 (0.47, 1.54)	-0.16 (-0.76, 0.43)	0.60
left_cheek	0.43 (0.27, 1.80)	-0.85 (-1.31, 0.59)	0.31 (0.13, 0.78)	-1.16 (-2.08, -0.25)	0.01
right_cheek	0.71 (0.41, 1.46)	-0.34 (-0.89, 0.38)	0.52 (0.06, 4.18)	-0.65 (-2.73, 1.43)	0.54
left_temple	0.43 (0.19, 0.73)	-0.85 (-1.64, -0.32)	0.35 (0.19, 0.65)	-1.06 (-1.68, -0.44)	0.00
right_temple	0.73 (0.49, 2.18)	-0.32 (-0.71, 0.78)	0.70 (0.39, 1.26)	-0.36 (-0.95, 0.23)	0.23
forehead	0.63 (0.50, 0.88)	-0.47 (-0.70, -0.12)	0.57 (0.38, 0.84)	-0.57 (-0.96, -0.18)	0.00
left_cheekbone	1.65 (1.05, 2.29)	0.50 (0.04, 0.83)	2.08 (1.23, 3.52)	0.73 (0.21, 1.26)	0.01
right_cheekbone	1.44 (0.62, 2.96)	0.36 (-0.48, 1.09)	1.90 (0.21, 16.95)	0.64 (-1.55, 2.83)	0.57
nose	0.99 (0.65, 1.57)	-0.01 (-0.43, 0.45)	0.91 (0.53, 1.55)	-0.09 (-0.63, 0.44)	0.73
left_eyebrow	1.49 (0.97, 2.09)	0.40 (-0.03, 0.74)	1.59 (1.02, 2.49)	0.47 (0.02, 0.91)	0.04
right_eyebrow	1.84 (0.72, 4.12)	0.61 (-0.34, 1.41)	1.99 (0.65, 6.08)	0.69 (-0.43, 1.80)	0.23
left_jaw	1.77 (1.21, 2.58)	0.57 (0.19, 0.95)	2.09 (1.18, 3.70)	0.74 (0.17, 1.31)	0.01
right_jaw	1.00 (0.43, 1.46)	-0.00 (-0.84, 0.38)	1.02 (0.59, 1.78)	0.02 (-0.53, 0.58)	0.93

Table A.5: Combined Results of Multivariate L2-Penalised Logistic Regression and GEE Sensitivity Analysis for Grad-CAM (Phase 2).

A.3 Model and Data Cards

Component	Essentials	Governance, risks & access
MorphoXAI (Explainer)	Overview: Post-hoc projection of CAMs onto a fixed facial-ROI ontology to yield a per-image <i>semantic attribution vector</i>. Intended use: Audit/interpretation; cohort-level statistics of attention distribution. Dependencies: Detection/landmarking (Sec. 2.1.2), stable ROI taxonomy, consistent pre-processing. Limitations: Constrained to predefined ROIs; may miss extra-facial cues.	Calibration: N/A (explainer). Maintenance: Re-validate ROI histograms and calibration after model/data changes; log drift. Ethics: Avoid sensitive-attribute inference; ensure subgroup-consistent pre-processing.
Face2Gender (Classifier)	Overview: Binary CNN on UTKFace to validate MorphoXAI at scale. Pre-proc/train: Sec. 2.2; backbone and training in Sec. 2.2.3. Intended use: Methodological validation/stress-testing only (no deployment).	Calibration & threshold: 0.5 operating point for selection; metrics on held-out split (Sec. 3.1.1). Risks: Filename label noise; demographic imbalance; context leakage. Maintenance: Not applicable (research artefact).
DBS Treatment-Response (Classifier)	Overview: Frame-based classifier (PD pre-/post-DBS) to quantify peri-oral/ocular redistribution. Pre-proc/train: Sec. 2.3; intended for research use, not diagnostic.	Calibration & threshold: Temperature scaling on validation; operating threshold τ^* chosen on calibrated scores (Sec. 2.3.5; reported in Table A.2). Risks: Small sample; recording-condition heterogeneity; limited external validity. Maintenance: Site-specific re-validation prior to any clinical study.
UTKFace (Dataset)	Summary: In-the-wild faces with filename-encoded attributes; used in Phase 1 (Sec. 2.2.1). Processing: Detection/alignment, resize, normalisation, splits as in Sec. 2.2.	Access/licensing: Public research distribution; verify licence before redistribution. Ethics: Potential demographic bias; sensitive-attribute handling; re-identification risk. Governance: Suitable for benchmarking/method validation.
DBS Video Cohort (Dataset)	Summary: Video frames of PD patients pre-/post-DBS; patient-disjoint splits (Sec. 2.3.1). Processing: Detection, alignment, grayscale→RGB replication, normalisation; frame sampling (Sec. 2.3.2).	Provenance/consent: Public YouTube sources used strictly for research; no redistribution of raw media; derived, de-identified features only under agreement. Ethics: Clinical data sensitivity; secure storage; subgroup fairness checks; document site/population limits.

Table A.6: Consolidated Model & Data Cards for MorphoXAI and associated components.

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